Latent Variable Moderated Mediation Using Correct and Misspecified Models

Felipe Fontana Vieira and Yves Rosseel

Department of Data Analysis, Faculty of Psychology and Educational Sciences, Ghent University

Author Note

Felipe Fontana Vieira https://orcid.org/0009-0006-0949-6569

Yves Rosseel https://orcid.org/0000-0002-4129-4477

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Correspondence concerning this article should be addressed to Felipe Fontana Vieira, Department of Data Analysis, Faculty of Psychology and Educational Sciences, Ghent University, Henri Dunantlaan 2, Ghent 9000, Belgium, Email: felipe.fontanavieira@ugent.be

Abstract

Moderated mediation is seldom estimated with latent variables, and its estimators have rarely been compared under model misspecification. In a preregistered simulation we evaluated three estimation approaches of a latent moderated mediation model, the local structural-after-measurement approach (LSAM) and two system-wide methods, across sample size, interaction magnitude, reliability, predictor distribution, and nine (mis)specification conditions. Under correct specification, all three recovered the interaction coefficient and index of moderated mediation, differing modestly in bias, mean squared error, coverage, standard errors, Type I error, and power. Under misspecification, LSAM was most robust, leaving some estimates unaffected by omissions that biased the system-wide estimators, although all misspecifications degraded performance.

Keywords: moderated mediation, structural equation modeling, latent interaction, measurement error, model misspecification

Latent Variable Moderated Mediation Using Correct and Misspecified Models

Mediation analysis is a well-known procedure in the behavioral and social sciences (Charlton et al., 2021; Hayes, 2018; Pieters, 2017; Preacher et al., 2007). In its basic and standard form, the effect of an independent variable on an outcome is decomposed into a direct effect and an indirect effect transmitted through a mediator, with all variables observed. This single mediator model has since been extended in a number of directions, most relevant here being the case in which the indirect effect is allowed to depend on a moderator (Edwards & Lambert, 2007; Hayes, 2015; MacKinnon et al., 2007; Montoya, 2024; Preacher et al., 2007).

In this context, a fundamental concern may arise from measurement. Many constructs under investigation are not directly observed but are measured indirectly through multiple items, that is, they are latent variables (Bollen & Hoyle, 2012). When such constructs are reduced to summary scores (e.g., a sum or mean of the items) and analyzed as though they were observed, a restrictive measurement structure is imposed rather than tested (McNeish & Wolf, 2020) and measurement error is left unmodelled. This error can attenuate or inflate the indirect effect and distort inferences about it (Cole & Preacher, 2014; Hoyle & Kenny, 1999), and it is especially consequential for the mediator, which acts at once as a predictor and an outcome (Gonzalez & MacKinnon, 2021). The problem is compounded under moderation, where the product term is even less reliable than either of its constituents (Bohrnstedt & Marwell, 1978; Busemeyer & Jones, 1983). To address this, structural equation modelling (SEM) with latent variables has been advocated, especially in the context of mediation (Cheung & Lau, 2008; Falk & Biesanz, 2015; Ledgerwood & Shrout, 2011; Montoya, 2024). It estimates structural relationships between common factors, from which indicator-specific unique variance has been partitioned, provided the measurement model is correctly specified (Bollen, 1989; DeShon, 1998; Rhemtulla et al., 2020). Yet we also note that latent variables do not necessarily improve power in general, whether for interactions or not (Cham et al., 2012; Fuller, 1987; Ledgerwood & Shrout, 2011; Wang & Rhemtulla, 2021).

Moderated mediation with latent variables is not straightforward in terms of statistical estimation. Indeed, a latent interaction[1] involves the product of two latent variables, which is unobserved and nonnormal even when its constituents are normal (Kelava & Brandt, 2022). Following this, a number of methods have been proposed to estimate nonlinear effects among latent variables (e.g., Algina & Moulder, 2001; Arminger & Muthén, 1998; Asparouhov & Muthén, 2021; Boker et al., 2023; Bollen, 1995; Jöreskog & Yan, 1996; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Marsh et al., 2004; Mooijaart & Bentler, 2010; Mooijaart & Satorra, 2012; Wall & Amemiya, 2000). Most adopt a joint ("system-wide") estimation strategy, in which the measurement and structural parameters are estimated simultaneously. This is the case of the known methods: latent moderated structural equations (LMS) and product-indicator (PI). Opposed to these are structural-after-measurement (SAM) approaches (Cox & Kelcey, 2021; Holst & Budtz-Jørgensen, 2020; Ng & Chan, 2020; Rosseel et al., 2025; Wall & Amemiya, 2001, 2003), in which the measurement model is estimated first and the structural model afterward with the measurement parameters held fixed. Many of these approaches have so far been rarely applied in practice, likely held back by limited software, slow estimation, and error-prone implementations. With recent open-source implementations (Holst & Budtz-Jørgensen, 2020; Rosseel et al., 2025; Slupphaug et al., 2024) and non-commercial software (Keller, 2025), this is likely to change.

Beyond estimation, a further consideration concerns interpretation. Mediation is a hypothesis about causal effects, and causal interpretation requires identification assumptions that *may* not be verified from the data alone (Imai et al., 2010; Morelli et al., 2026; Rohrer et al., 2022; VanderWeele, 2015). With latent variable models, further caution is needed (Sterner et al., 2024; VanderWeele & Vansteelandt, 2022). We focus on one that is central to interpretation: the estimated latent variables must validly represent the intended constructs (Gonzalez & MacKinnon, 2021). In fact, when they do not, the model is subject to interpretational confounding

[1] We use interaction and moderation interchangeably, as the focus is on estimation, but we highlight they imply different interpretations and causal assumptions (VanderWeele, 2009, 2015). An interaction also need not be a product term, although that is the most common functional form used by researchers (Rimpler et al., 2026).

(Burt, 1976), a discrepancy between the empirical meaning assigned to a latent variable through its estimated measurement parameters and the meaning intended by the researcher. This reflects an underlying property of system-wide estimation, namely that the consequences of a misspecification are not confined to the part of the model in which it occurs (Bainter & Bollen, 2014). Burt (1976) proposed estimating the measurement model first, as a SAM approach does (e.g., Rosseel & Loh, 2024), so that the estimated latent variables and their meaning do not depend on the structural model that follows, although this does not imply that the structure is correct (VanderWeele & Vansteelandt, 2022).

To illustrate interpretational confounding, we take a mediation model with three latent variables, X, M, and Y, each with three indicators, and obtain a population covariance matrix from a correctly specified version. We then alter it so that the covariances between the X and M indicators are unequal across the X indicators (3, 1, and 1, rather than the implied value of 2), so their association with M no longer follows a single common factor. To the same data we fit two structural models, one omitting the mediator and one including it, by system-wide SEM and by SAM (N = 200), and compare the estimated loadings of X. Without the mediator, both approaches recover the true loadings of one. Once the mediator enters, the system-wide loadings of x_2 and x_3 drop to 0.41, even though the measurement of X is unchanged, whereas the SAM loadings remain at one with and without the mediator.

Despite this, prior simulation work has examined latent interaction estimation mainly under distributional misspecification (Brandt et al., 2014; e.g., Cham et al., 2012; Lonati et al., 2024; Vieira et al., 2026), and in only two cases under misspecification at the measurement level (Brandt et al., 2020; Navarro & Schermelleh-Engel, 2025), which between them considered only the UPI and LMS among the approaches studied here in limited settings. Moreover, when it comes to (moderated) mediation, despite its popularity in applied research, only a few studies have investigated it with latent variables, and those assumed a correctly specified (and saturated) model (Cheung & Lau, 2017; Falk & Biesanz, 2015; Feng et al., 2020). In this article, we therefore evaluate three estimation approaches for a latent variable moderated mediation model:

local SAM (LSAM; Rosseel et al., 2025), LMS (Kelava et al., 2011; Klein & Moosbrugger, 2000), and the unconstrained PI approach (UPI; Kelava & Brandt, 2009; Marsh et al., 2004). These approaches have been explained extensively in previous research and we refer readers to it (e.g., Brandt et al., 2020; Kelava et al., 2011; Lin et al., 2010; Lonati et al., 2024; Rosseel et al., 2025; Slupphaug et al., 2024; Vieira et al., 2026).

Present Research and Contributions

This study makes two contributions. First, we evaluate LSAM alongside LMS and UPI under both measurement and structural misspecification, in a considerably broader design than previous studies, detailed in the next section. The robustness of LSAM to model misspecification has itself been examined only in linear models (Rosseel & Loh, 2024), with later research disputing it for measurement misspecifications and expecting it to hold mainly for structural misspecifications in non-saturated structural models (Robitzsch, 2022), a case not yet tested for models with nonlinear effects. Second, we obtain inference for the index of moderated mediation (imm; Hayes, 2015), a nonlinear function of the structural parameters, with the Monte Carlo method (MacKinnon et al., 2004; Preacher & Selig, 2012) rather than the bootstrap used in the latent variable studies above (Cheung & Lau, 2017; Feng et al., 2020). Bootstrapping is computationally expensive, and its variants differ modestly in performance and rank differently depending on the criterion (Falk & Biesanz, 2015; Tibbe & Montoya, 2022), whereas the Monte Carlo method needs only the estimates and their covariance matrix. Whether it performs adequately for the imm in the context under investigation, which has not been examined, is a question we address here.

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate performance, following some recent recommendations for the conduct and reporting of simulation studies (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024).

Population Model and Estimands

We study a moderated mediation model among four latent variables: an exogenous predictor X, a moderator W, a mediator M, and an outcome Y. The structural model is as follows

$$M = a_1 X + a_2 W + a_3\,(X \times W) + \zeta_M, \qquad Y = b\,M + c'\,X + \zeta_Y,$$

The primary estimands are the interaction coefficient (a_3) and the imm (i.e., $a_3 b$). The remaining structural coefficients (a_1, a_2, b, c') and the twelve freely estimated factor loadings serve as secondary estimands. Note the imm is a derived quantity rather than a free parameter, hence its true value is computed as $a_3 b$ implied by the correctly specified model.

Population values were fixed at $a_1 = 0.4$, $a_2 = 0.3$, $b = 0.5$, and $c' = 0.15$, with $\mathrm{cor}(X, W) = 0.3$. The interaction coefficient a_3 is a design factor (see below). These coefficients, together with the structural disturbance variances, were chosen so that under the baseline condition (normal exogenous predictors and a correctly specified model) each dependent latent variable has $R^2 = 0.30$ at the interaction level $a_3 = 0.20$. Also note that, because a_3 contributes to the variance of M, the explained variance is naturally higher with a larger a_3 and lower when its effect is null. Each latent variable is measured by four congeneric indicators with loadings 1.0, 0.8, 0.7, and 0.6. Figure 1 shows the population model, where the measurement and structural misspecifications studied below are drawn in grey.

Simulation Conditions

We employed a fully factorial design crossing five factors: sample size ($N \in \{100, 200, 300, 500\}$), interaction magnitude ($a_3 \in \{0, 0.20, 0.40\}$), measurement reliability (low = 0.5, high = 0.7), the distribution of the exogenous latent predictors (normal, uniform, t, and two χ^2 conditions with same- or opposite-direction skew), and the (mis)specification conditions (nine cases; see Data Generation below). This yielded $4 \times 3 \times 2 \times 5 \times 9 = 1{,}080$ conditions, each replicated 1000 times.

Sample sizes cover the small to moderate range, where interaction terms are estimated with limited power (Brandt et al., 2020; Lonati et al., 2024). The interaction magnitudes include a null condition to assess Type I error for the interaction coefficient and imm, and two nonzero levels representing small and moderate latent moderation effects. Reliability levels of 0.5 and 0.7 were chosen because measurement error is amplified in product terms and is therefore especially consequential for latent interaction estimation (Brandt et al., 2020; Lonati et al., 2024; Vieira et al., 2026). Nonnormality of the latent predictors is both empirically common and methodologically central for latent interaction estimators, whose distributional assumptions differ across approaches (Cham et al., 2012; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Lonati et al., 2024; Vieira et al., 2026). The misspecification conditions follow the moderated mediation framework for the two structural omissions (Preacher et al., 2007) and prior latent interaction simulation work for the two measurement omissions (Brandt et al., 2020). They were also chosen as interpretable violations, acting respectively as a measured mediator–outcome confounder (VanderWeele, 2015), a conflated moderated direct effect (Edwards & Lambert, 2007), exposure-correlated measurement error in the mediator (Fritz et al., 2016; VanderWeele et al., 2012), and a shared unmeasured cause of a mediator and an outcome indicator (Cole et al., 2007). The misspecification cases are detailed below.

Estimation Approaches

LSAM (Rosseel et al., 2025; Rosseel & Loh, 2024), with the measurement model estimated by ML in the first step. In this case, the interacting predictors X and W shared a single measurement block, while M and Y formed their own blocks. LMS (Klein & Moosbrugger, 2000), with robust standard errors, and the extended UPI approach with double mean centering (Kelava & Brandt, 2009; Lin et al., 2010; Marsh et al., 2004), referred to as UPI throughout, estimated with robust maximum likelihood (MLR), that is, ML with Huber–White sandwich standard errors.

Data Generation

Appendix describes the complete data-generating mechanism in detail. Briefly, the latent variables X and W were drawn from a bivariate distribution with $\text{Cor}(X, W) = \rho = 0.3$, constructed with the vine-to-anything (VITA) algorithm (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022). Five exogenous distributions were examined, each standardized to mean

0 and variance 1. Thus, the conditions differ only in distributional shape, not in scale or location. Specifically, the normal condition draws $X, W \sim \mathcal{N}(0, 1)$ through a forced Gaussian copula, whereas the remaining four combine non-normal margins with a non-Gaussian copula: uniform, t_5, and two $\chi^2(2)$ conditions in which X and W are skewed in the same or in opposite directions. Given that, M and Y were generated from the structural equations above with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$, and the four indicators ($j = 1, \ldots, 4$) of each latent variable $\eta \in \{X, W, M, Y\}$ were formed as $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$, with loadings $\lambda = (1, 0.8, 0.7, 0.6)$ and Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$ whose variances $\theta_{\eta j}$ were scaled to the target reliability $\omega \in \{0.5, 0.7\}$.

The nine (mis)specification conditions comprise one correctly specified condition and four types of misspecification, each at two empirically calibrated magnitudes (standardized 0.25 and 0.50, detailed in Appendix), similar to Brandt et al. (2020). Two are structural: an omitted direct effect of the moderator on the outcome (c_2) and an omitted moderated direct effect (c_3). The others are in the measurement model: a cross-loading of a mediator indicator on X (`cl`) and a residual covariance between a mediator and an outcome indicator (`rc`). In every case the effect is present in the population but omitted from the estimated model.

Convergence, Outliers, and Inadmissible Solutions

Following previous research (Lonati et al., 2024; Vieira et al., 2026), replications were screened before analysis in two sequential steps. In both, a replication flagged for any one estimator was removed for all three, so that every estimator is compared on an identical set of replications. First, we checked convergence. A fit was treated as nonconverged when any estimate, standard error, or confidence-interval bound of any structural coefficient or loading was missing or non-finite, and a replication was removed from all estimators whenever any one of them did not converge. Second, among the remaining replications, we screened for outliers, identified per condition, method, and parameter as point estimates beyond the interquartile range ± 3, together with replications whose estimated standard error exceeded 10 on any parameter of interest[2]. Again, a replication flagged as an outlier for any estimation approach was removed from all of them. We also flagged inadmissible solutions, where any residual (unique) variance was nonpositive or the covariance matrix of the exogenous latent variables and latent disturbances was not positive definite.

Performance Measures

Performance was assessed with the measures implemented in the *simhelpers* package (Joshi & Pustejovsky, 2025), computed for the imm, the structural coefficients, and the loadings. These measures, with their empirical and Monte Carlo standard error (MCSE) calculations, are listed in Table 1. Note the MCSE are obtained in closed form where available and by jackknife otherwise. On top of that, where applicable, we also report the rejection rate (Type I error when $a_3 = 0$, power otherwise).

Inference for the imm followed the Monte Carlo method as mentioned before: for $r = 1, \ldots, R$ (with $R = 20{,}000$) we drew

$$\big(a_3^{(r)}, b^{(r)}\big) \sim \mathcal{N}_2\big((\hat{a}_3, \hat{b}),\ \hat{\Sigma}_{a_3 b}\big), \qquad \mathrm{imm}^{(r)} = a_3^{(r)}\, b^{(r)},$$

where $\hat{\Sigma}_{a_3 b}$ is the estimated covariance of $(\hat{a}_3, \hat{b})$, and took the 2.5th and 97.5th empirical percentiles of $\{\mathrm{imm}^{(r)}\}_{r=1}^{R}$ as the interval. The structural coefficients and loadings use Wald-based intervals from the same estimated covariance matrix of the parameter estimates.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2025) using the *lavaan* (Rosseel, 2012) and *modsem* (Slupphaug et al., 2024) packages for estimation, *covsim* (Grønneberg et al., 2022) and *rvinecopulib* (Nagler & Vatter, 2025) for data generation, and *simhelpers* (Joshi & Pustejovsky, 2025) for performance evaluation. Conditions were run in parallel with the *parallel*

[2] We added the standard error criterion because some replications retained anomalous standard errors despite acceptable point estimates. It additionally removed 125 replications (about 0.01% of the globally converged ones), predominantly for UPI at the smallest sample size with low reliability and skewed χ^2 predictors.

package. Reproducible and independent random-number streams across workers were obtained with the L'Ecuyer-CMRG generator. The computational environment was pinned with Nix (Dolstra et al., 2004) via the *rix* package (Rodrigues & Baumann, 2026) to ensure reproducibility, and the article was written with the *apaquarto* extension (Schneider, 2026). For the R version, package versions, code, manuscript, and materials, see the project repository.

Results

We present the results in three sections. First, the results for the interaction coefficient and the imm are reported. We then turn to the remaining structural coefficients (i.e., a_1, a_2, b, and c'), and we close with the estimated factor loadings, focusing mainly on that of the cross-loaded indicator (m_3). Moreover, due to the large number of conditions, we focused on a subset of the conditions. Namely, we fix the interaction coefficient at 0.2 in most cases (except for Type I error, which by construction requires a zero interaction) and display a limited number of sample sizes: $N = 200$ and 500 for the interaction coefficient and imm, and solely $N = 500$ for the structural coefficients and loadings. For the latter two, we further restrict to high reliability, whereas the interaction coefficient and imm are shown at both reliability levels. The entire set of results as well as detailed information about convergence and outliers is available in the repository.

Convergence, Outliers, and Inadmissible Solutions

Convergence was high throughout. LSAM converged in every replication across all conditions, and LMS in at least 99.9%. UPI was the least stable, with its convergence rate falling to about 92% in the worst condition (the smallest sample size, low reliability, and skewed χ^2 predictors). At high reliability, however, UPI converged in at least 99% of replications regardless of sample size, distribution, or misspecification. Outliers were likewise rare, staying below 1% in the majority of conditions and reaching their highest values, up to roughly 11% for UPI and 8% for LSAM, only at $N = 100$ with low reliability and skewed (χ^2) predictors. Note the results reported below did not exclude inadmissible solutions, which arose almost exclusively for UPI. In fact, excluding them leads to negligible changes, with the few differences confined to the smallest sample size, low reliability, χ^2 conditions.

Main Structural Estimates

Relative Bias

We interpret values whose absolute relative bias exceeds 0.10 (10%) as biased, following previous research (Brandt et al., 2014; Vieira et al., 2026). Under correct specification with normal predictors, all three estimators recover both the interaction coefficient and imm without bias at every sample size and reliability level, and the same holds under uniform and t predictors apart from single LMS cells at $N = 100$ with low reliability (Figure 2 and Figure 3). Moreover, the χ^2 conditions affect all three approaches, but differently. UPI and LSAM are biased only at $N = 100$ with low reliability, where UPI shows the largest deviation, whereas LMS remains biased across all sample sizes at low reliability, negatively under opposite skew at the larger interaction and positively under same skew at the smaller one.

Under misspecification, both c_2 and c_3 bias the imm for all three estimation approaches across distributions, as expected. The size of the bias nevertheless varies with the predictor distribution, being largest under normal and same-skew χ^2 predictors and smallest under opposite-skew χ^2 and t_5 predictors. For the interaction coefficient, c_2 produces essentially no bias for all approaches, whereas c_3 leads to bias under the system-wide approaches. Among the measurement omissions, `cl` leaves the imm unbiased except for LMS in the χ^2 conditions, and attenuates the interaction coefficient only under the system-wide methods. Last, `rc` induces an upward bias in the imm at low reliability, most pronounced for LMS under same-skew χ^2, and leaves the interaction coefficient largely unchanged. For the structural omissions, the bias increases with the magnitude of the omission in every condition, and it is generally larger at low reliability (Figure 4). We therefore also note that the bias of the interaction coefficient under c_3 is the one result not implied by the omission alone.

Relative RMSE

Under correct specification with normal predictors, the relative RMSE of both the interaction coefficient and the imm falls with sample size and reliability for all three estimators, with LMS the most efficient, LSAM intermediate, and UPI the least (Figure 5). In general, the same holds for both quantities under the nonnormal predictors. Under misspecification, the structural omissions (c_2 and c_3) inflate the RMSE of the imm for all three estimators. Under c_3 the inflation is clearly larger for the system-wide estimators, especially at low reliability, whereas under c_2 the three are more similar. For the interaction coefficient, c_3 inflates the RMSE for both system-wide estimators, more so at low reliability, while LSAM is unaffected. The measurement omissions (`cl` and `rc`) have a negligible effect, except for LMS under the medium `cl` at low reliability.

Standard errors

We interpret SE/SD ratios below 0.90 (underestimation) or above 1.10 (overestimation) as biased (Brandt et al., 2014; Vieira et al., 2026), and similarly for the relative bias of the variance estimator. Note that the standard errors for the imm are based on the Monte Carlo method.

Under correct specification with normal predictors, the SE/SD ratio of both the interaction coefficient and the imm is unbiased for all three approaches, except for UPI, which overestimates at $N = 200$ with low reliability (Figure 6 and Figure 7). Under the nonnormal predictors, that overestimation extends to both χ^2 conditions for UPI and, under opposite-skew χ^2, to LSAM, while at $N = 500$ only UPI under opposite-skew χ^2 remains above the band. In addition, under t_5 predictors all three approaches marginally underestimate the standard error of the interaction coefficient at high reliability. Misspecification barely alters this pattern. The same deviations recur throughout all omissions, with only scattered additional cells. The relative bias of the variance estimator amplifies the same pattern. At high reliability, UPI under both χ^2 conditions and LSAM under opposite-skew χ^2 exceed the band across most (mis)specifications, with LMS deviating only in scattered cells, and at low reliability UPI overestimates the variance under every distribution, LSAM mainly under opposite-skew χ^2, and LMS under opposite-skew χ^2 and uniform predictors (figures available in the project repository).

Coverage

Coverage between 92.5% and 97.5% was considered acceptable. Under correct specification with normal predictors, coverage is around the nominal level with all three approaches (Figure 8 and Figure 9) for the interaction coefficient and imm. Still with a correct specification, it dips slightly below the band under heavy-tailed (t_5) predictors for all three approaches at both reliability levels, and under same-skew χ^2 predictors at low reliability, most for LMS at $N = 500$. Under misspecification, c_2 and c_3 reduce the imm coverage, since its interval is then centered on a biased estimate, with the largest drop for the system-wide estimators under c_3. The drop grows with sample size and is present under most distributions, but it is less intense under opposite-skew χ^2 and, at $N = 200$ with low reliability, largely absent for UPI and LSAM. For the interaction coefficient, the medium cross-loading (`cl`) is the most damaging case, lowering coverage under the system-wide estimators, most for LMS at low reliability, while LSAM holds near nominal. c_3 also lowers it under the system-wide estimators, again most for LMS at low reliability, whereas c_2 and `rc` leave coverage close to nominal.

Hypothesis Testing (Type I error and Power)

Under correct specification with normal predictors and $a_3 = 0$, the interaction coefficient and imm hold their nominal level reasonably well for LSAM and LMS, while UPI is conservative at $N = 200$ with low reliability (Figure 10). Under the nonnormal predictors, LMS shows inflated Type I error under t_5 at both reliability levels and under both χ^2 conditions at low reliability, LSAM under t_5 at high reliability, whereas UPI is conservative under opposite-skew χ^2 and uniform predictors at low reliability. Under misspecification with normal predictors, only the medium c_3 inflates the false-positive rate of the system-wide estimators, most for LMS at low reliability. With nonnormal predictors, the correctly specified pattern persists across most (mis)specifications, with LMS inflated under t_5 and both χ^2 distributions, LSAM and UPI inflated under t_5 in a number of cells, and UPI conservative under opposite-skew χ^2.

At $a_3 = 0.2$ the imm is detected with power that rises with sample size and reliability under normal predictors, with LMS the most powerful (Figure 11). Under the nonnormal predictors, LMS keeps the highest power under χ^2, whereas under t_5 and uniform predictors LSAM matches or slightly exceeds it, and power falls substantially under opposite-skew χ^2 predictors, most for UPI and LSAM. We note, however, that interpreting power in the case of bias and inflated Type I error should be avoided (Morris et al., 2019).

Additional Structural Estimates

For the remaining structural coefficients (a_1, a_2, b, and c'), at high reliability and $N = 500$, we report their relative bias (Figure 12), coverage (Figure 13), and SE/SD ratio (Figure 14) across all misspecification conditions and distributions. Under correct specification, all approaches produced unbiased results, acceptable coverage and SE/SD ratios throughout distributions. With misspecifications, however, for the relative bias, the two structural omissions act on the outcome equation: c_2 biases the mediator-outcome slope b and, more strongly, the direct effect c', and c_3 is absorbed mainly by b and c' as well, for all three estimators alike. Under LSAM the mediator-equation coefficients a_1 and a_2 stay essentially unbiased under both structural omissions, since LSAM estimates the mediator block in isolation. The system-wide estimators, by contrast, propagate c_2 into the mediator equation, biasing a_2, whereas c_3 leaves a_1 and a_2 unaffected for all three and surfaces instead in the interaction coefficient, as reported above. The cross-loading (`cl`) is the most disruptive, biasing a_1 and c' for every estimator and, under the system-wide estimators, a_2 and b as well, because the omitted X loading on m_3 distorts the mediator's relations to the other latent variables. The residual covariance (`rc`) slightly inflates b and biases c' downward for all three, within the threshold. Where the bias is substantial, it is smaller under LSAM than under the system-wide estimators, except for c' under the structural omissions, where the three are alike. Moreover, coverage follows this bias pattern: near nominal where bias is small and dropping where bias is large, most sharply for b under c_2, for c' under c_3, and for a_1 under the cross-loading. The SE/SD ratio stays close to one for all three estimators, with one clear exception: for b and c' under c_3 the standard errors are underestimated, most pronounced under heavy-tailed t_5 predictors.

Measurement Loadings

For the twelve freely estimated factor loadings, all but one are recovered with near-nominal performance across every measure and condition. Hence, we display only the cross-loaded mediator indicator m_3, with its relative bias, coverage, SE/SD ratio, and relative bias of the variance estimator shown together at high reliability and $N = 500$ (Figure 15). For m_3, the omitted cross-loading (`cl`) is the only consequential misspecification: its relative bias rises sharply, to about half at the medium level under the system-wide estimators and about a third less under LSAM, and its coverage collapses toward zero. The two standard error measures, by contrast, stay within the desirable range, except for LSAM under the medium cross-loading with t_5 and χ^2 predictors, where both fall below the band. We also highlight that a subtler pattern, negligible in size, appears under the residual covariance (`rc`): the system-wide estimators carry a small systematic upward bias in m_3, under one percent, whereas LSAM stays centered on zero.

Discussion

We evaluated three estimation approaches in the context of a latent variable moderated mediation model, the two-step LSAM and the system-wide LMS and UPI, under correct specification and under a large number of structural and measurement misspecifications as well as distributional violations.

Under correct specification, all three recovered the imm and the interaction coefficient with negligible relative bias and near-nominal coverage, differing mainly in terms of coverage, Type I error, and the SE/SD ratio on one side, and relative RMSE and power on the other. LMS had the lowest relative RMSE and the highest power but showed below-nominal coverage and inflated Type I error, increasingly so under nonnormal predictors. LSAM yielded coverage and Type I error closest to nominal, at the cost of a higher relative RMSE and lower power in some cases, while UPI fell in between, accurate in relative bias but the least stable at the smallest sample size and low reliability. The two standard error performance measures, the SE/SD ratio and the relative bias of the variance estimator, showed similar results: both stayed near one at high reliability but not at low reliability and small samples. Indeed, UPI overestimated standard errors, and UPI and LSAM struggled the most under the χ^2 predictors skewed in opposite directions.

Under misspecification, LSAM was the more robust approach. The bias from an omission was more limited in LSAM than under the system-wide estimators: the structural omissions in the outcome equation (c_2, c_3) biased the outcome-side coefficients but left the mediator equation, and hence the interaction coefficient, unaffected, whereas LMS and UPI also biased the interaction coefficient. The omitted cross-loading (`cl`) was the most damaging throughout, biasing every coefficient and sharply lowering the coverage of the interaction coefficient under the system-wide estimators at low reliability, while LSAM remained near nominal; the residual covariance (`rc`) had little effect. Across conditions the relative bias was consistently smaller under LSAM, and it increased with the magnitude of the omission and at lower reliability. This pattern matches the expectation of Robitzsch (2022) that the advantage of SAM lies in structural rather than measurement misspecification, since the cross-loading biased LSAM as well, whereas the structural omissions left its mediator equation untouched.

These patterns are broadly consistent with previous research. The results for the LMS and UPI approaches under specific types of nonnormality alongside its stability under normality reproduces earlier findings (Brandt et al., 2014; Cham et al., 2012; Lonati et al., 2024; Vieira et al., 2026) and the behavior of LSAM also mirrors previous studies (Rosseel et al., 2025; Vieira et al., 2026). Moreover, part of measurement misspecifications results is aligned with Brandt et al. (2020). There, omitting a cross-loading was likewise a damaging misspecification, and the UPI approach was biased under it, whereas the two-stage method of moments on which LSAM builds attained the smallest RMSE in those conditions. Navarro and Schermelleh-Engel (2025) likewise found the latent interaction effect underestimated under omitted cross-loadings with LMS, which they trace to the biased predictor covariance entering the variance of the interaction term. Even so, that study found no estimator free of bias and none uniformly best, as we also observed.

Finally, although this article evaluated the estimation approaches from a statistical, inferential perspective, the moderated mediation model itself rests on strong causal assumptions (VanderWeele, 2015). This carries a practical consequence for applied researchers: good statistical performance of the estimation approaches under investigation in certain conditions is not, on its own, a justification for a causal research question or claim. Work addressing the causal interpretation of mediation continues to develop, and we direct interested readers to this literature (Imai et al., 2010; Morelli et al., 2026; Rohrer et al., 2022). Moreover, as we showed, misspecification has strong consequences across all three estimation approaches, and many of those may be highly plausible in certain context and fields.

References

Algina, J., & Moulder, B. C. (2001). A note on estimating the Jöreskog–Yang model for latent variable interaction using LISREL 8.3. *Structural Equation Modeling: A Multidisciplinary Journal*, *8*(1), 40–52. https://doi.org/10.1207/S15328007SEM0801_3

Arminger, G., & Muthén, B. O. (1998). A Bayesian approach to nonlinear latent variable models using the Gibbs sampler and the Metropolis–Hastings algorithm. *Psychometrika*, *63*(3), 271–300. https://doi.org/10.1007/bf02294856

Asparouhov, T., & Muthén, B. (2021). Bayesian estimation of single and multilevel models with latent variable interactions. *Structural Equation Modeling: A Multidisciplinary Journal*, *28*(2), 314–328. https://doi.org/10.1080/10705511.2020.1761808

Bainter, S. A., & Bollen, K. A. (2014). Interpretational confounding or confounded interpretations of causal indicators? *Measurement: Interdisciplinary Research and Perspectives*, *12*(4), 125–140. https://doi.org/10.1080/15366367.2014.968503

Bohrnstedt, G. W., & Marwell, G. (1978). The reliability of products of two random variables. *Sociological Methodology*, *9*, 254–273. https://doi.org/10.2307/270812

Boker, S. M., Oertzen, T. von, Pritikin, J. N., Hunter, M. D., Brick, T. R., Brandmaier, A. M., & Neale, M. C. (2023). Products of variables in structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *30*(5), 708–718. https://doi.org/10.1080/10705511.2022.2141749

Bollen, K. A. (1989). *Structural equations with latent variables*. John Wiley & Sons. https://doi.org/10.1002/9781118619179

Bollen, K. A. (1995). Structural equation models that are nonlinear in latent variables: A least-squares estimator. *Sociological Methodology*, *25*, 223–251. https://doi.org/10.2307/271068

Bollen, K. A., & Hoyle, R. H. (2012). Latent variables in structural equation modeling. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (pp. 56–67). The Guilford Press.

Brandt, H., Kelava, A., & Klein, A. G. (2014). A simulation study comparing recent approaches for the estimation of nonlinear effects in SEM under the condition of nonnormality. *Structural Equation Modeling: A Multidisciplinary Journal*, *21*(2), 181–195. https://doi.org/10.1080/10705511.2014.882660

Brandt, H., Umbach, N., Kelava, A., & Bollen, K. A. (2020). Comparing estimators for latent interaction models under structural and distributional misspecifications. *Psychological Methods*, *25*(3), 321–345. https://doi.org/10.1037/met0000231

Burt, R. S. (1976). Interpretational confounding of unobserved variables in structural equation models. *Sociological Methods & Research*, *5*(1), 3–52. https://doi.org/10.1177/004912417600500101

Busemeyer, J. R., & Jones, L. E. (1983). Analysis of multiplicative combination rules when the causal variables are measured with error. *Psychological Bulletin*, *93*(3), 549–562. https://doi.org/10.1037/0033-2909.93.3.549

Cham, H., West, S. G., Ma, Y., & Aiken, L. S. (2012). Estimating latent variable interactions with nonnormal observed data: A comparison of four approaches. *Multivariate Behavioral Research*, *47*(6), 840–876. https://doi.org/10.1080/00273171.2012.732901

Charlton, A., Montoya, A. K., Price, J., Hilgard, J., & Krefeld-Schwalb, A. (2021). *Noise in the process: An assessment of the evidential value of mediation effects in marketing journals*. PsyArXiv preprint. https://doi.org/10.31234/osf.io/ck2r5

Cheung, G. W., & Lau, R. S. (2008). Testing mediation and suppression effects of latent variables: Bootstrapping with structural equation models. *Organizational Research Methods*, *11*(2), 296–325. https://doi.org/10.1177/1094428107300343

Cheung, G. W., & Lau, R. S. (2017). Accuracy of parameter estimates and confidence intervals in moderated mediation models: A comparison of regression and latent moderated structural equations. *Organizational Research Methods*, *20*(4), 746–769. https://doi.org/10.1177/1094428115595869

Cole, D. A., Ciesla, J. A., & Steiger, J. H. (2007). The insidious effects of failing to include design-driven correlated residuals in latent-variable covariance structure analysis.

Psychological Methods, *12*(4), 381–398. https://doi.org/10.1037/1082-989X.12.4.381

Cole, D. A., & Preacher, K. J. (2014). Manifest variable path analysis: Potentially serious and misleading consequences due to uncorrected measurement error. *Psychological Methods*, *19*(2), 300–315. https://doi.org/10.1037/a0033805

Cox, K., & Kelcey, B. (2021). Croon's bias-corrected estimation of latent interactions. *Structural Equation Modeling: A Multidisciplinary Journal*, *28*(6), 863–874. https://doi.org/10.1080/10705511.2021.1922283

DeShon, R. P. (1998). A cautionary note on measurement error corrections in structural equation models. *Psychological Methods*, *3*(4), 412–423. https://doi.org/10.1037/1082-989X.3.4.412

Dolstra, E., De Jonge, M., & Visser, E. (2004). Nix: A safe and policy-free system for software deployment. *18th Large Installation System Administration Conference*, 79–92.

Edwards, J. R., & Lambert, L. S. (2007). Methods for integrating moderation and mediation: A general analytical framework using moderated path analysis. *Psychological Methods*, *12*(1), 1–22. https://doi.org/10.1037/1082-989X.12.1.1

Falk, C. F., & Biesanz, J. C. (2015). Inference and interval estimation methods for indirect effects with latent variable models. *Structural Equation Modeling: A Multidisciplinary Journal*, *22*(1), 24–38. https://doi.org/10.1080/10705511.2014.935266

Feng, Q., Song, Q., Zhang, L., Zheng, S., & Pan, J. (2020). Integration of moderation and mediation in a latent variable framework: A comparison of estimation approaches for the second-stage moderated mediation model. *Frontiers in Psychology*, *11*, 2167. https://doi.org/10.3389/fpsyg.2020.02167

Foldnes, N., & Grønneberg, S. (2015). How general is the Vale–Maurelli simulation approach? *Psychometrika*, *80*(4), 1066–1083. https://doi.org/10.1007/s11336-014-9414-0

Fritz, M. S., Kenny, D. A., & MacKinnon, D. P. (2016). The combined effects of measurement error and omitting confounders in the single-mediator model. *Multivariate Behavioral Research*, *51*(5), 681–697. https://doi.org/10.1080/00273171.2016.1224154

Fuller, W. A. (1987). *Measurement error models*. Wiley. https://doi.org/10.1002/9780470316665

Gonzalez, O., & MacKinnon, D. P. (2021). The measurement of the mediator and its influence on statistical mediation conclusions. *Psychological Methods*, *26*(1), 1–17. https://doi.org/10.1037/met0000263

Grønneberg, S., Foldnes, N., & Marcoulides, K. M. (2022). covsim: An R package for simulating non-normal data for structural equation models using copulas. *Journal of Statistical Software*, *102*(3), 1–45. https://doi.org/10.18637/jss.v102.i03

Hayes, A. F. (2015). An index and test of linear moderated mediation. *Multivariate Behavioral Research*, *50*(1), 1–22. https://doi.org/10.1080/00273171.2014.962683

Hayes, A. F. (2018). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach* (2nd ed.). The Guilford Press.

Holst, K. K., & Budtz-Jørgensen, E. (2020). A two-stage estimation procedure for non-linear structural equation models. *Biostatistics*, *21*(4), 676–691. https://doi.org/10.1093/biostatistics/kxy082

Hoyle, R. H., & Kenny, D. A. (1999). Sample size, reliability, and tests of statistical mediation. In R. H. Hoyle (Ed.), *Statistical strategies for small sample research* (pp. 195–222). Sage.

Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, *15*(4), 309–334. https://doi.org/10.1037/a0020761

Jöreskog, K. G., & Yan, F. (1996). Nonlinear structural equation models: The Kenny–Judd model with interaction effects. In G. A. Marcoulides & R. E. Schumacker (Eds.), *Advanced structural equation modeling* (pp. 57–88). Lawrence Erlbaum Associates.

Joshi, M., & Pustejovsky, J. (2025). *simhelpers: Helper functions for simulation studies*. https://CRAN.R-project.org/package=simhelpers

Kelava, A., & Brandt, H. (2009). Estimation of nonlinear latent structural equation models using the extended unconstrained approach. *Review of Psychology*, *16*(2), 123–131.

Kelava, A., & Brandt, H. (2022). Latent interaction effects. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (pp. 427–446). Guilford Press.

Kelava, A., Werner, C. S., Schermelleh-Engel, K., Moosbrugger, H., Zapf, D., Ma, Y., Cham, H., Aiken, L. S., & West, S. G. (2011). Advanced nonlinear latent variable modeling: Distribution analytic LMS and QML estimators of interaction and quadratic effects. *Structural Equation Modeling: A Multidisciplinary Journal*, *18*(3), 465–491. https://doi.org/10.1080/10705511.2011.582408

Keller, B. T. (2025). A factored regression approach to modeling latent variable interactions and nonlinear effects. *Psychological Methods*. https://doi.org/10.1037/met0000777

Kenny, D. A., & Judd, C. M. (1984). Estimating the nonlinear and interactive effects of latent variables. *Psychological Bulletin*, *96*(1), 201–210. https://doi.org/10.1037/0033-2909.96.1.201

Klein, A. G., & Moosbrugger, H. (2000). Maximum likelihood estimation of latent interaction effects with the LMS method. *Psychometrika*, *65*(4), 457–474. https://doi.org/10.1007/BF02296338

Klein, A. G., & Muthén, B. O. (2007). Quasi-maximum likelihood estimation of structural equation models with multiple interaction and quadratic effects. *Multivariate Behavioral Research*, *42*(4), 647–673.

Ledgerwood, A., & Shrout, P. E. (2011). The trade-off between accuracy and precision in latent variable models of mediation processes. *Journal of Personality and Social Psychology*, *101*(6), 1174–1188. https://doi.org/10.1037/a0024776

Lin, G.-C., Wen, Z., Marsh, H. W., & Lin, H.-S. (2010). Structural equation models of latent interactions: Clarification of orthogonalizing and double-mean-centering strategies. *Structural Equation Modeling: A Multidisciplinary Journal*, *17*(3), 374–391. https://doi.org/10.1080/10705511.2010.488999

Lonati, S., Rönkkö, M., & Antonakis, J. (2024). Normality assumption in latent interaction models. *Psychological Methods*. https://doi.org/10.1037/met0000657

MacKinnon, D. P., Fairchild, A. J., & Fritz, M. S. (2007). Mediation analysis. *Annual Review of Psychology*, *58*, 593–614. https://doi.org/10.1146/annurev.psych.58.110405.085542

MacKinnon, D. P., Lockwood, C. M., & Williams, J. (2004). Confidence limits for the indirect effect: Distribution of the product and resampling methods. *Multivariate Behavioral Research*, *39*(1), 99–128. https://doi.org/10.1207/s15327906mbr3901_4

Marsh, H. W., Wen, Z., & Hau, K.-T. (2004). Structural equation models of latent interactions: Evaluation of alternative estimation strategies and indicator construction. *Psychological Methods*, *9*(3), 275–300. https://doi.org/10.1037/1082-989X.9.3.275

McNeish, D., & Wolf, M. G. (2020). Thinking twice about sum scores. *Behavior Research Methods*, *52*(6), 2287–2305. https://doi.org/10.3758/s13428-020-01398-0

Montoya, A. K. (2024). Combining statistical and causal mediation analysis. In H. T. Reis, T. West, & C. M. Judd (Eds.), *Handbook of research methods in social and personality psychology* (pp. 622–652). Cambridge University Press.

Mooijaart, A., & Bentler, P. M. (2010). An alternative approach for nonlinear latent variable models. *Structural Equation Modeling: A Multidisciplinary Journal*, *17*(3), 357–373. https://doi.org/10.1080/10705511.2010.488997

Mooijaart, A., & Satorra, A. (2012). Moment testing for interaction terms in structural equation modeling. *Psychometrika*, *77*(1), 65–84. https://doi.org/10.1007/s11336-011-9232-6

Morelli, S., Faleh, R., & Brandt, H. (2026). *RAPSEM: Identifying latent mediators without sequential ignorability via a rank-preserving structural equation model*. arXiv preprint arXiv:2509.23935. https://doi.org/10.48550/arXiv.2509.23935

Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, *38*(11), 2074–2102. https://doi.org/10.1002/sim.8086

Nagler, T., & Vatter, T. (2025). *rvinecopulib: High performance algorithms for vine copula modeling*. https://CRAN.R-project.org/package=rvinecopulib

Navarro, K., & Schermelleh-Engel, K. (2025). Cause for concern: Omitted cross-loadings in measurement models of nonlinear structural equation models. *Behavior Research Methods*, *57*(328). https://doi.org/10.3758/s13428-025-02792-2

Ng, J. C. K., & Chan, W. (2020). Latent moderation analysis: A factor score approach. *Structural*

Equation Modeling: A Multidisciplinary Journal, *27*(4), 629–648. https://doi.org/10.1080/10705511.2019.1664304

Pawel, S., Bartoš, F., Siepe, B. S., & Lohmann, A. (2025). Handling missingness, failures, and non-convergence in simulation studies: A review of current practices and recommendations. *The American Statistician*, 1–18. https://doi.org/10.1080/00031305.2025.2540002

Pieters, R. (2017). Meaningful mediation analysis: Plausible causal inference and informative communication. *Journal of Consumer Research*, *44*(3), 692–716. https://doi.org/10.1093/jcr/ucx081

Preacher, K. J., Rucker, D. D., & Hayes, A. F. (2007). Addressing moderated mediation hypotheses: Theory, methods, and prescriptions. *Multivariate Behavioral Research*, *42*(1), 185–227. https://doi.org/10.1080/00273170701341316

Preacher, K. J., & Selig, J. P. (2012). Advantages of monte carlo confidence intervals for indirect effects. *Communication Methods and Measures*, *6*(2), 77–98. https://doi.org/10.1080/19312458.2012.679848

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Rhemtulla, M., Bork, R. van, & Borsboom, D. (2020). Worse than measurement error: Consequences of inappropriate latent variable measurement models. *Psychological Methods*, *25*(1), 30–45. https://doi.org/10.1037/met0000220

Rimpler, A., Kiers, H. A. L., & Ravenzwaaij, D. van. (2026). Anything goes: Statistical interactions without substantive theory. *Computational Brain & Behavior*. https://doi.org/10.1007/s42113-026-00305-8

Robitzsch, A. (2022). Comparing the robustness of the structural after measurement (SAM) approach to structural equation modeling (SEM) against local model misspecifications with alternative estimation approaches. *Stats*, *5*(3), 631–672. https://doi.org/10.3390/stats5030039

Rodrigues, B., & Baumann, P. (2026). *Rix: Reproducible data science environments with Nix*. https://CRAN.R-project.org/package=rix

Rohrer, J. M., Hünermund, P., Arslan, R. C., & Elson, M. (2022). That's a lot to process! Pitfalls of popular path models. *Advances in Methods and Practices in Psychological Science*, *5*(2), 1–14. https://doi.org/10.1177/25152459221095827

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. https://doi.org/10.18637/jss.v048.i02

Rosseel, Y., Burghgraeve, E., Loh, W. W., & Schermelleh-Engel, K. (2025). Structural after measurement (SAM) approaches for accommodating latent quadratic and interaction effects. *Behavior Research Methods*, *57*(4), 101. https://doi.org/10.3758/s13428-024-02532-y

Rosseel, Y., & Loh, W. W. (2024). A structural after measurement approach to structural equation modeling. *Psychological Methods*, *29*(3), 561–588. https://doi.org/10.1037/met0000503

Schneider, W. J. (2026). *apaquarto: A Quarto extension for creating APA style (7th edition) documents*. https://github.com/wjschne/apaquarto

Siepe, B. S., Bartoš, F., Morris, T. P., Boulesteix, A.-L., Heck, D. W., & Pawel, S. (2024). Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting. *Psychological Methods*. https://doi.org/10.1037/met0000695

Slupphaug, K. S., Mehmetoglu, M., & Mittner, M. (2024). modsem: An R package for estimating latent interactions and quadratic effects. *Structural Equation Modeling: A Multidisciplinary Journal*. https://doi.org/10.1080/10705511.2024.2417409

Sterner, P., Pargent, F., Deffner, D., & Goretzko, D. (2024). A causal framework for the comparability of latent variables. *Structural Equation Modeling: A Multidisciplinary Journal*, *31*(5), 747–758. https://doi.org/10.1080/10705511.2024.2339396

Tibbe, T. D., & Montoya, A. K. (2022). Correcting the bias correction for the bootstrap confidence interval in mediation analysis. *Frontiers in Psychology*, *13*, 810258. https://doi.org/10.3389/fpsyg.2022.810258

VanderWeele, T. J. (2009). On the distinction between interaction and effect modification. *Epidemiology*, *20*(6), 863–871. https://doi.org/10.1097/EDE.0b013e3181ba333c

VanderWeele, T. J. (2015). *Explanation in causal inference: Methods for mediation and interaction*. Oxford University Press.

VanderWeele, T. J., Valeri, L., & Ogburn, E. L. (2012). The role of measurement error and misclassification in mediation analysis. *Epidemiology*, *23*(4), 561–564. https://doi.org/10.1097/EDE.0b013e318258f5e4

VanderWeele, T. J., & Vansteelandt, S. (2022). A statistical test to reject the structural interpretation of a latent factor model. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *84*(5), 2032–2054. https://doi.org/10.1111/rssb.12555

Vieira, F. F., Slupphaug, K. S., & Rosseel, Y. (2026). *Evaluating local structural-after-measurement (LSAM) and traditional approaches for the estimation of complex nonlinear effects among latent variables*. PsyArXiv preprint. https://osf.io/preprints/psyarxiv/xk8um_v3

Wall, M. M., & Amemiya, Y. (2000). Estimation for polynomial structural equation models. *Journal of the American Statistical Association*, *95*(451), 929–940. https://doi.org/10.2307/2669475

Wall, M. M., & Amemiya, Y. (2001). Generalized appended product indicator procedure for nonlinear structural equation analysis. *Journal of Educational and Behavioral Statistics*, *26*(1), 1–29. https://doi.org/10.3102/10769986026001001

Wall, M. M., & Amemiya, Y. (2003). A method of moments technique for fitting interaction effects in structural equation models. *British Journal of Mathematical and Statistical Psychology*, *56*(1), 47–63. https://doi.org/10.1348/000711003321645331

Wang, Y. A., & Rhemtulla, M. (2021). Power analysis for parameter estimation in structural equation modeling: A discussion and tutorial. *Advances in Methods and Practices in Psychological Science*, *4*(1), 1–17. https://doi.org/10.1177/2515245920918253

Table 1

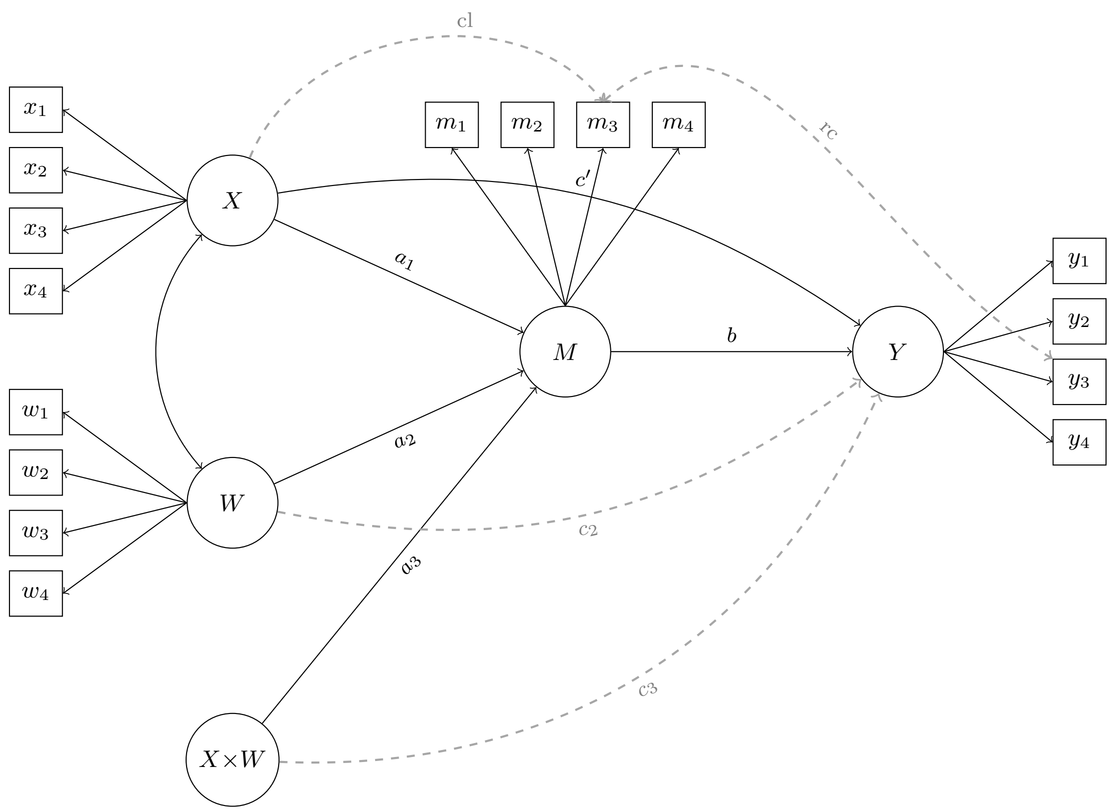
Subset of the Performance Measures for Evaluating Parameter Estimates

Performance measure	Empirical calculation	MCSE
Relative criteria		
Relative bias	$\frac{\bar{\hat{\theta}}}{\theta} - 1$	$\sqrt{S^2_{\hat{\theta}}/(K\theta^2)}$
Relative RMSE	$\frac{1}{\lvert\theta\rvert}\sqrt{\frac{1}{K}\sum_{k=1}^{K}(\hat{\theta}_k - \theta)^2}$	$\frac{1}{\lvert\theta\rvert}\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\mathrm{RMSE}_{(j)} - \mathrm{RMSE})^2}$
Standard errors		
Mean SE	$\overline{SE} = \frac{1}{K}\sum_{k=1}^{K}\widehat{SE}_k$	—
SE/SD ratio	$\overline{SE}/S_{\hat{\theta}}$	—
Relative bias of variance estimator	$\rho = \overline{SE^2}/S^2_{\hat{\theta}}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\rho_{(j)} - \rho)^2}$
Coverage		
Coverage	$c = \frac{1}{K}\sum_{k=1}^{K} I(L_k \leq \theta \leq U_k)$	$\sqrt{c(1-c)/K}$
Hypothesis testing		
Rejection rate	$r = \frac{1}{K}\sum_{k=1}^{K} I(0 \notin [L_k, U_k])$	$\sqrt{r(1-r)/K}$

Note. $\hat{\theta}_k$ is the estimate of parameter θ in replication k ($k = 1, \ldots, K$ usable replications), $\bar{\hat{\theta}}$ its mean, $S^2_{\hat{\theta}}$ its sample variance, and $S_{\hat{\theta}}$ its standard deviation; RMSE $= \sqrt{\frac{1}{K}\sum_k(\hat{\theta}_k - \theta)^2}$; $\widehat{SE}_k$ is the estimated standard error, $\overline{SE^2} = \frac{1}{K}\sum_k \widehat{SE}^2_k$, and L_k, U_k the confidence-interval limits; c and r denote coverage and rejection rate, and ρ the relative bias of the variance estimator. The subscript (j) marks a leave-one-out jackknife replicate. The MCSEs for relative RMSE and the relative bias of the variance estimator are computed by jackknife; the others are closed-form, and dashes mark MCSEs that are not computed.

Figure 1

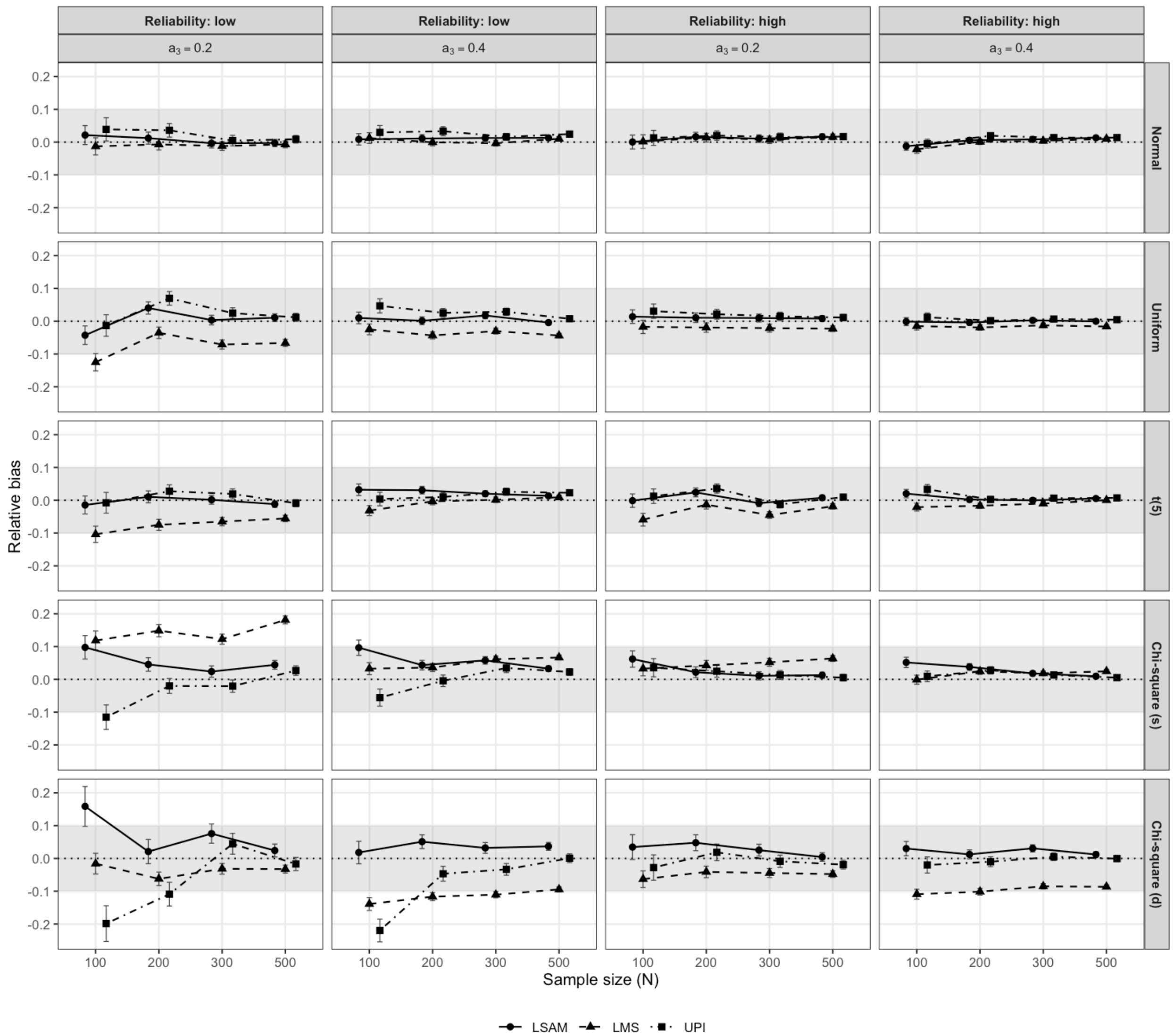
Population Model for the Simulation Study.



Note. X, W, M, and Y are latent variables, each measured by four congeneric indicators. Circles are latent variables and $X{\times}W$ is the latent interaction. Black elements form the correctly specified model that every estimator fits (a_1, a_2, a_3, b, c', and the X–W covariance). Grey elements are the misspecifications present in the population but omitted from the fitted model: two in the structural part, an omitted moderator effect on the outcome (c_2) and an omitted moderated direct effect (c_3); and two in the measurement part, an omitted cross-loading of X on m_3 (cl) and an omitted residual covariance between m_3 and y_3 (rc). Residual variances and intercepts are not shown.

Figure 2

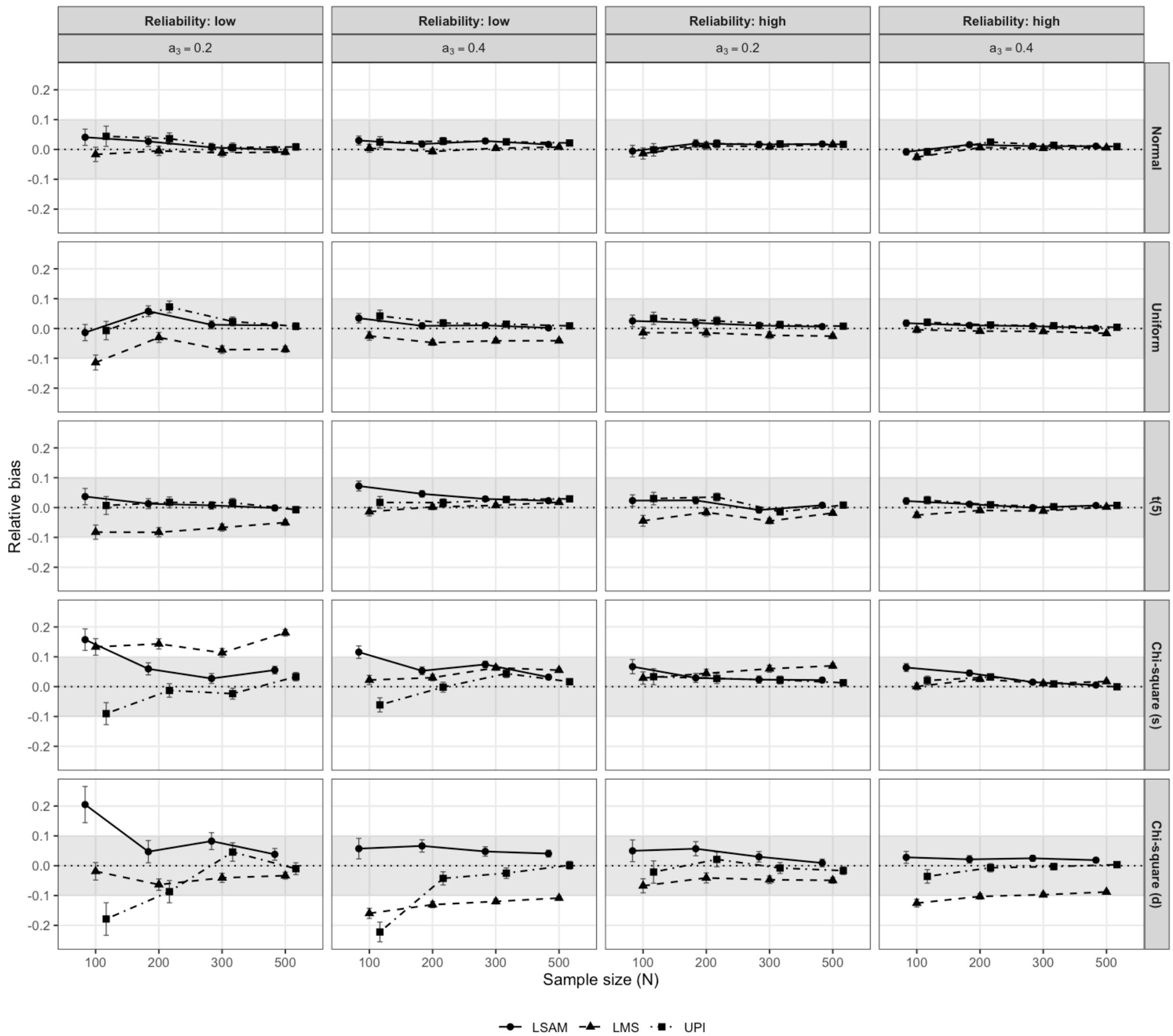
Relative Bias of the imm (a_3b) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 3

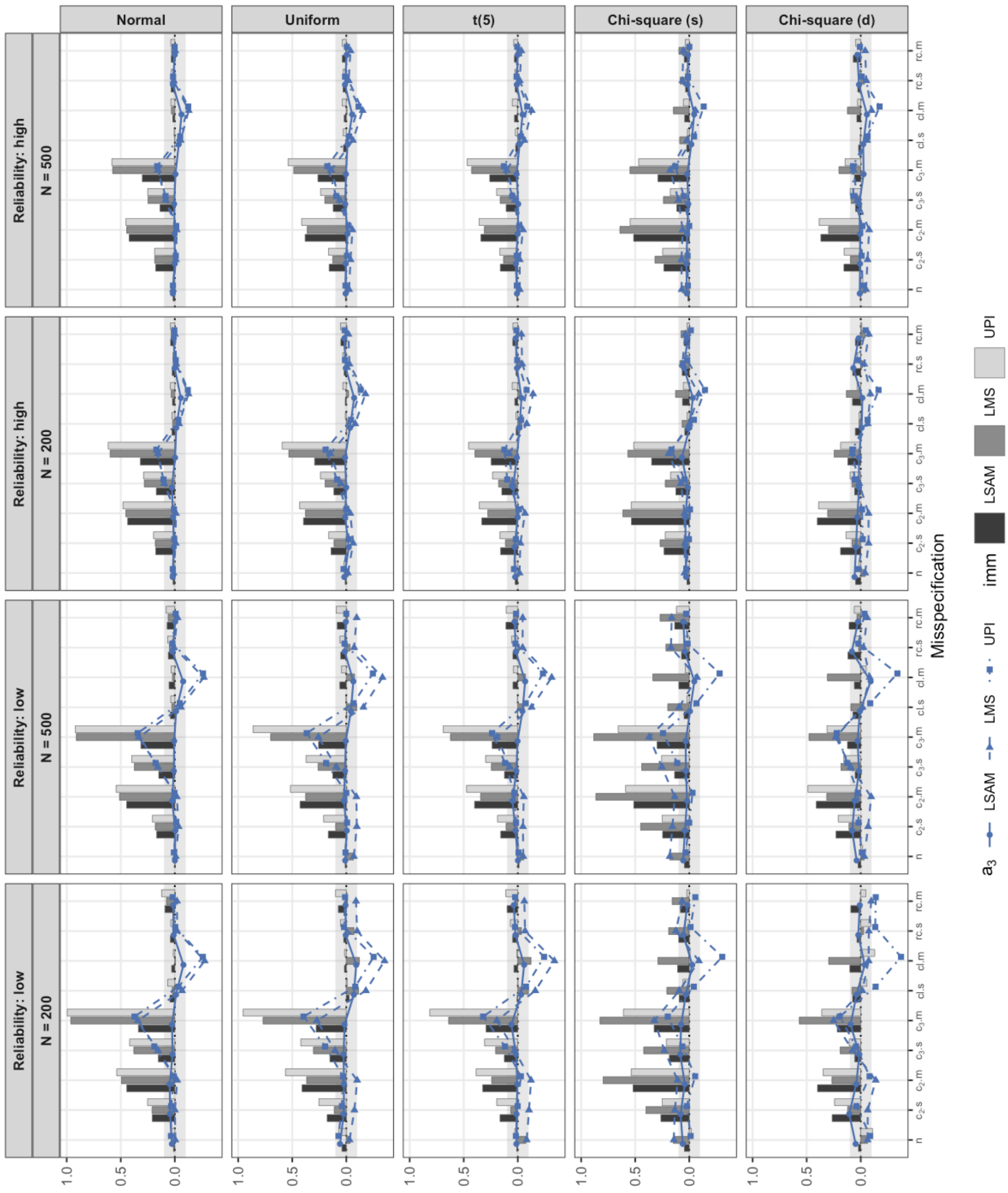
Relative Bias of the Interaction Coefficient (a_3) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 4

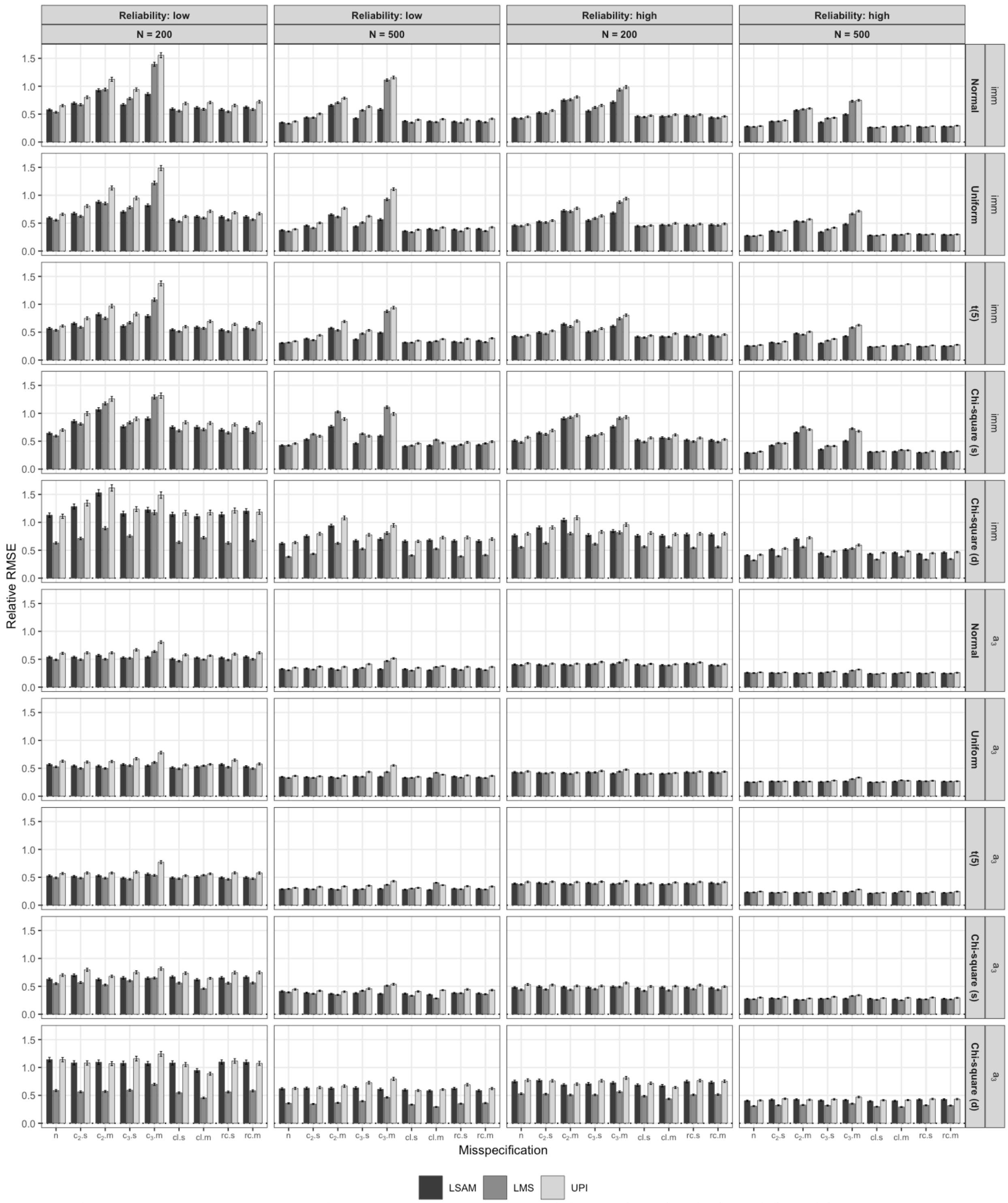
Relative Bias of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for relative bias. MCSEs are not shown.

Figure 5

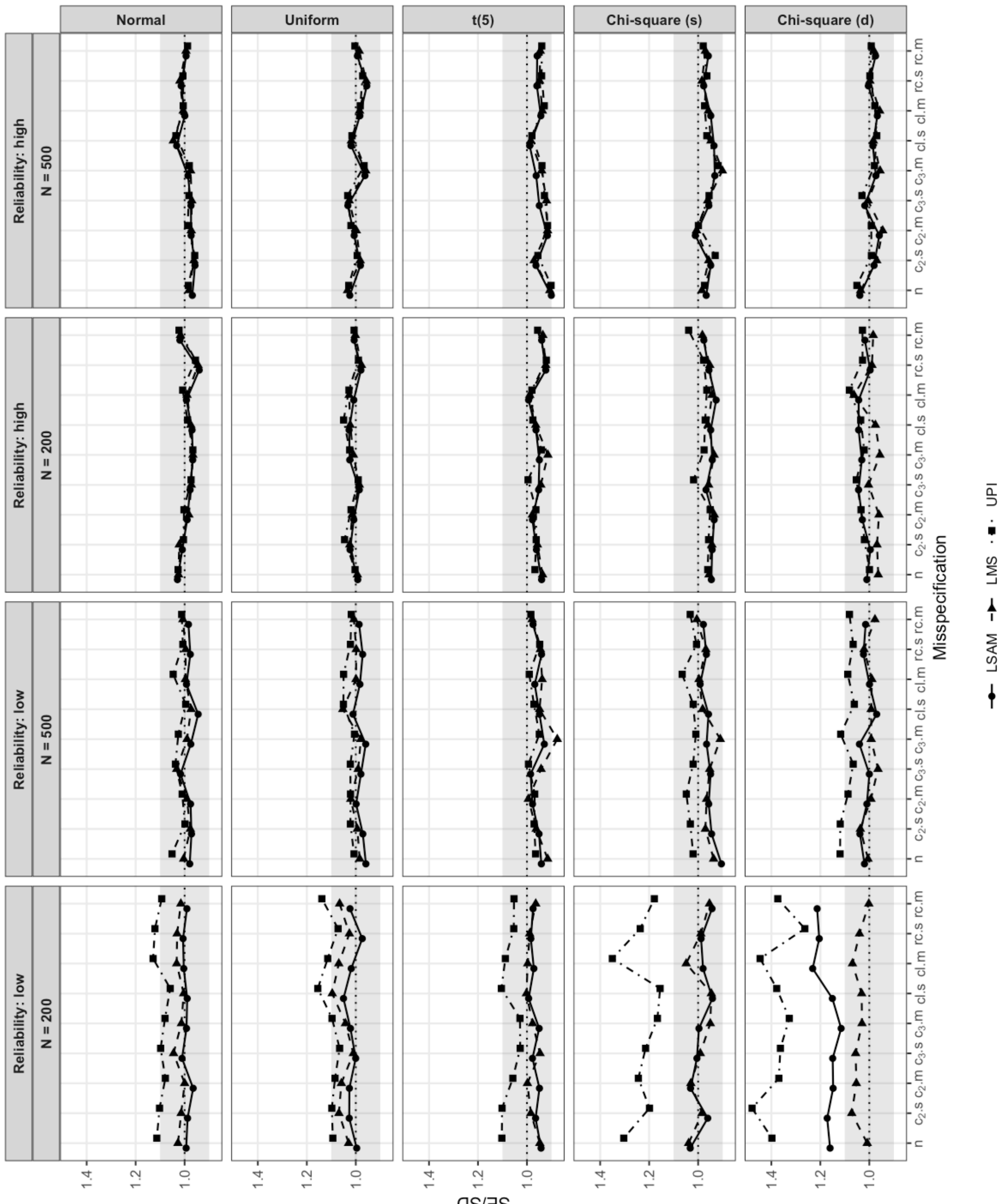
Relative RMSE of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The error bars represent the MCSE. No acceptable region is defined for this measure.

Figure 6

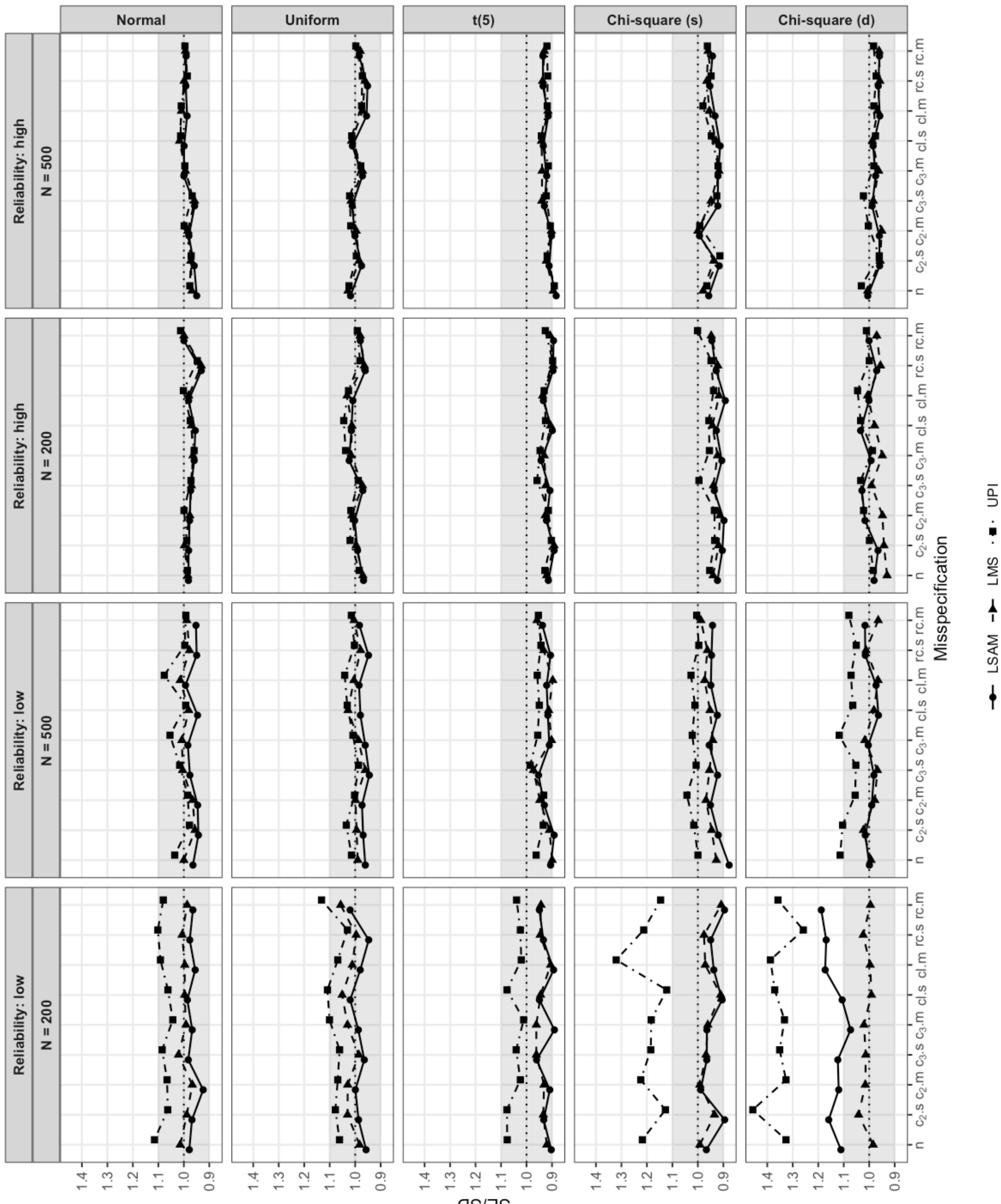
SE/SD of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for the measure. The MCSE is not computed for the SE/SD ratio.

Figure 7

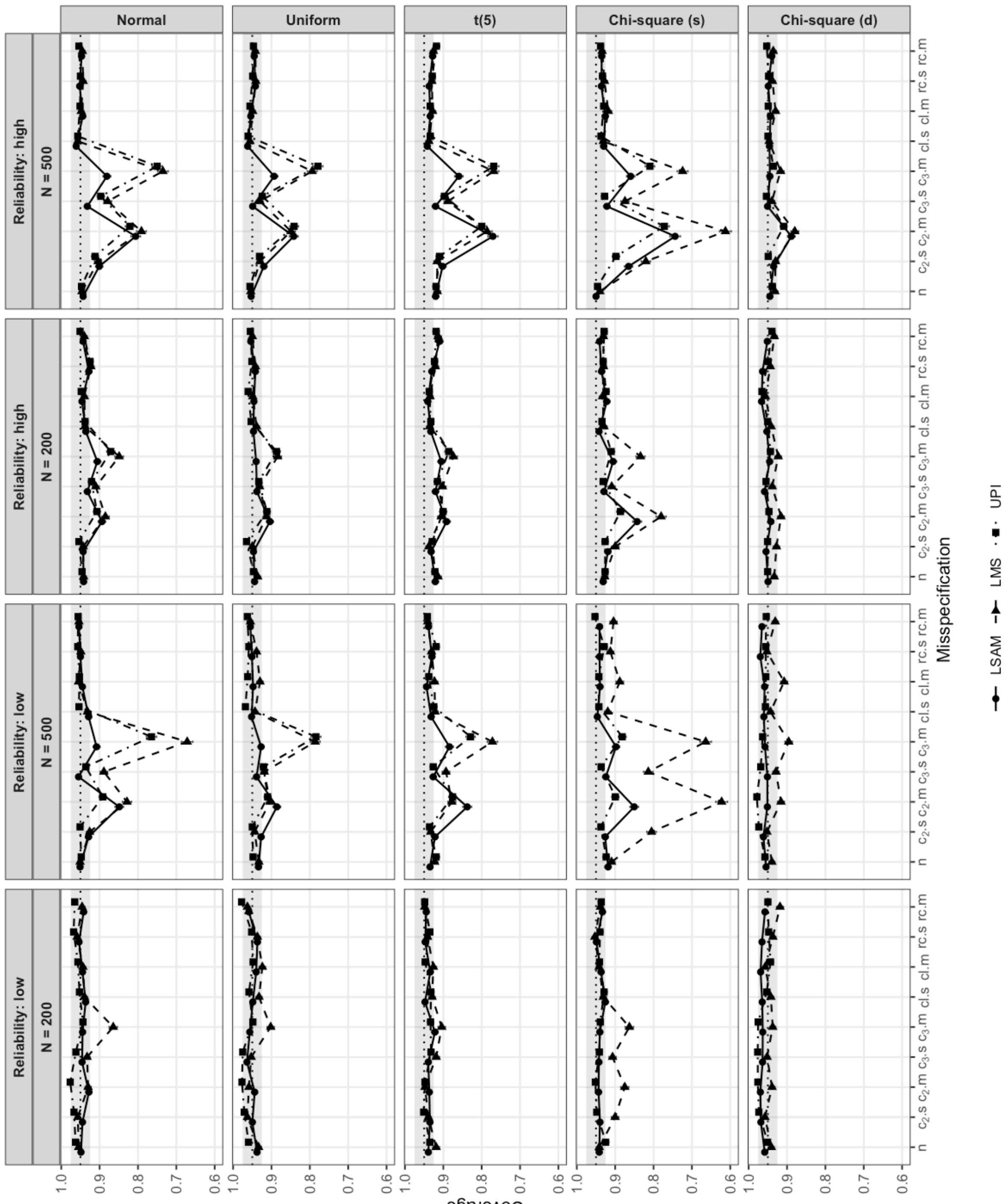
SE/SD of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for the measure. The MCSE is not computed for the SE/SD ratio.

Figure 8

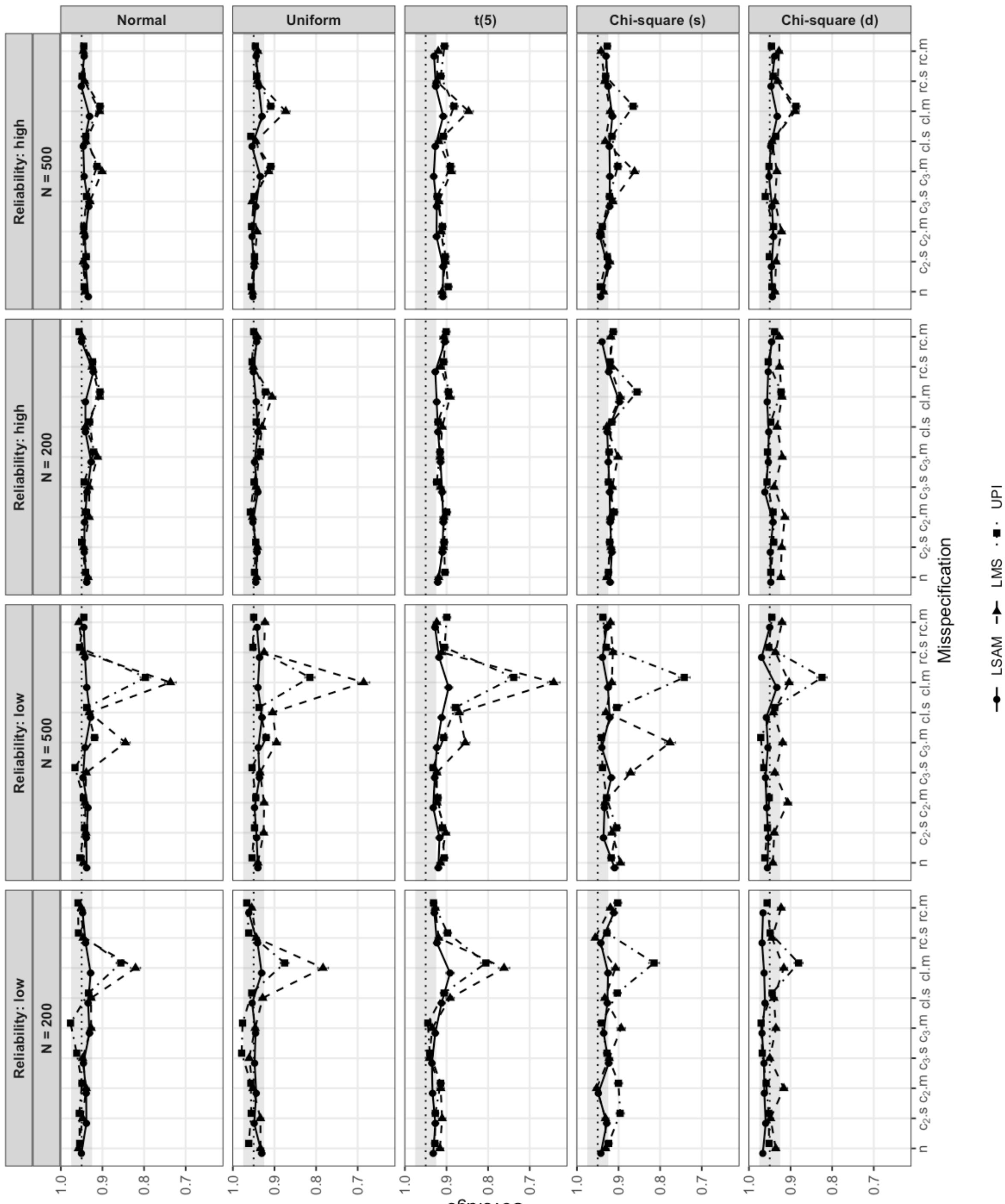
Coverage of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 9

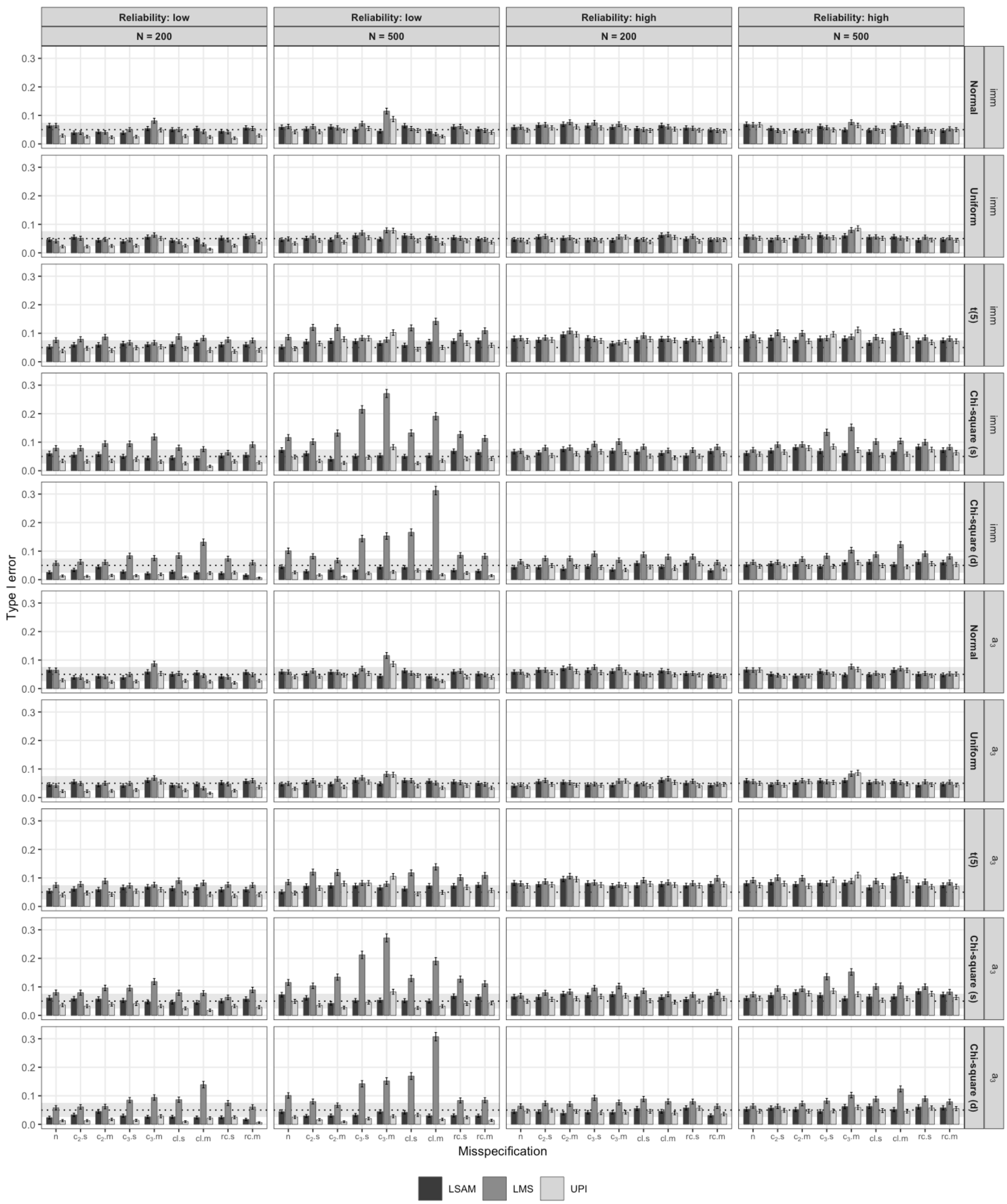
Coverage of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 10

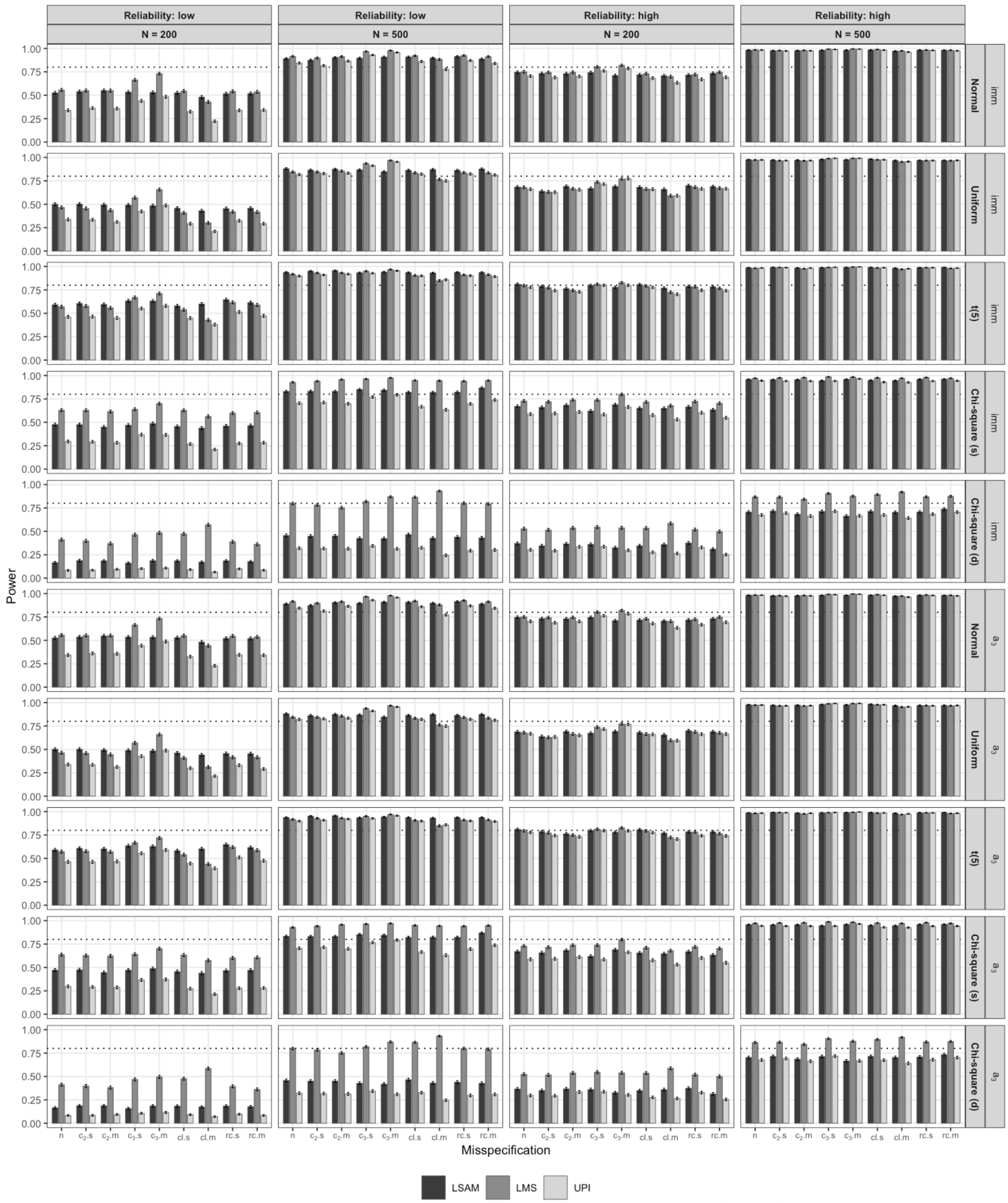
Type I Error of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0$



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 11

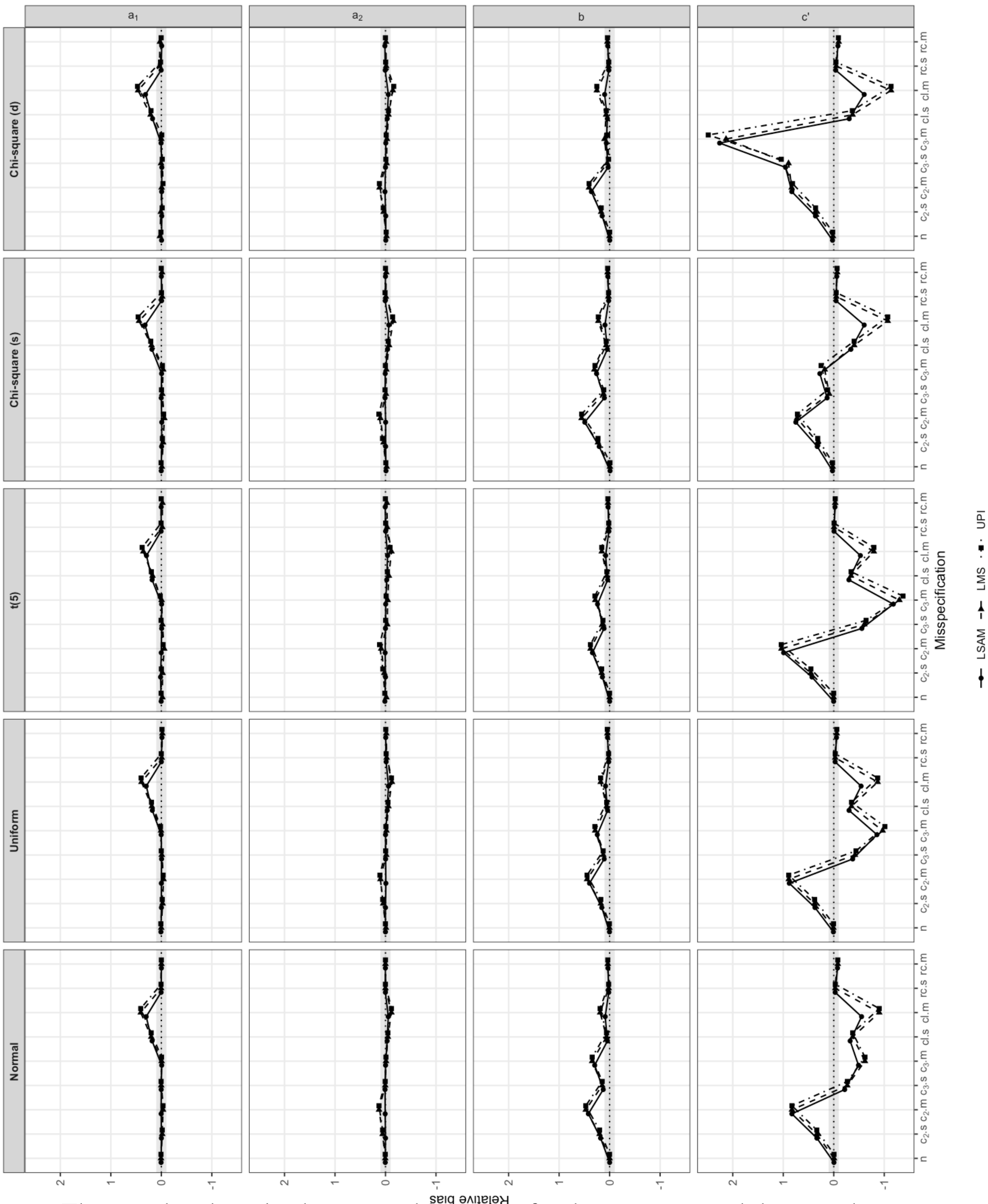
Power for the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The error bars represent the MCSE. No acceptable region is defined for this measure.

Figure 12

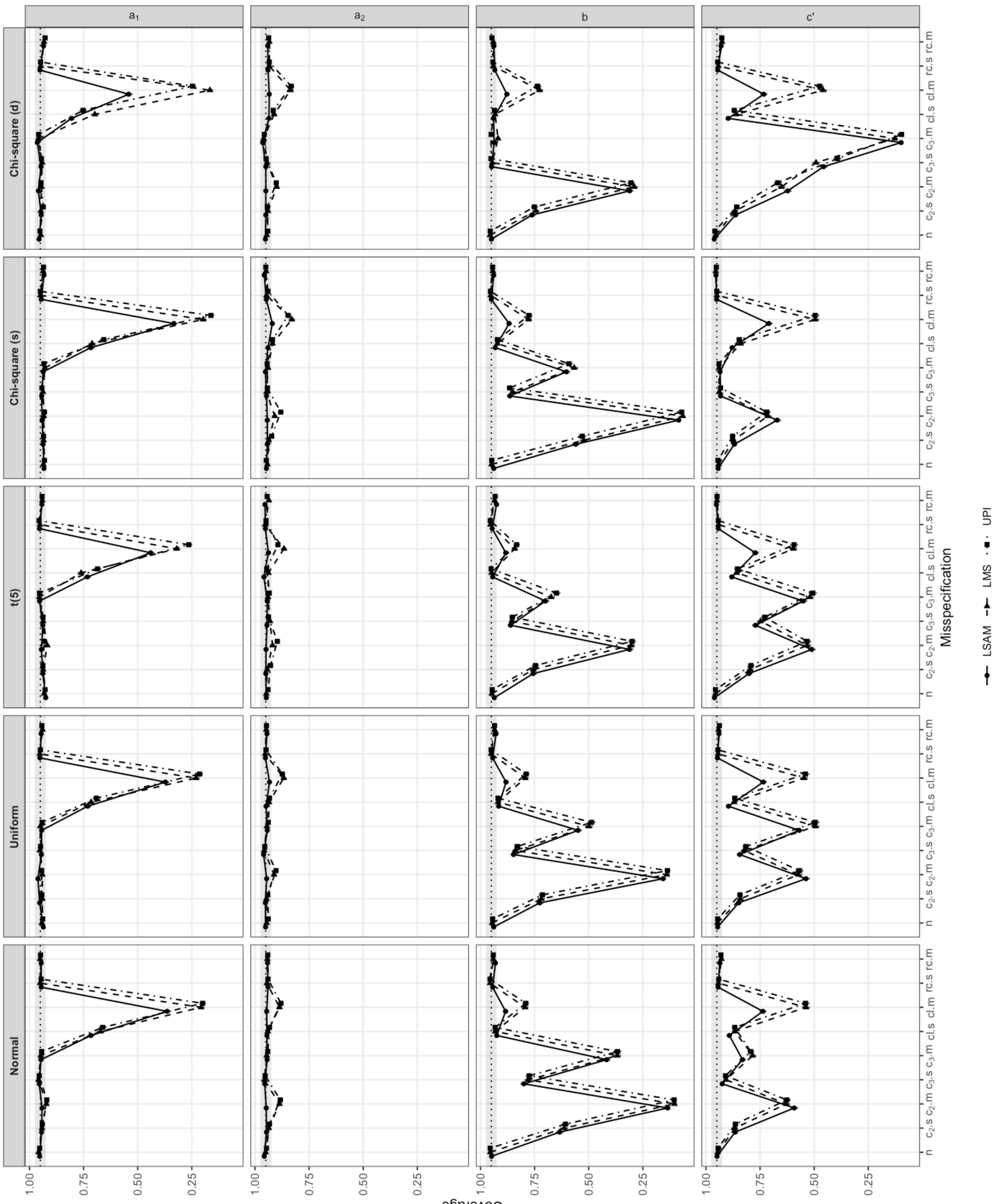
Relative Bias of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 13

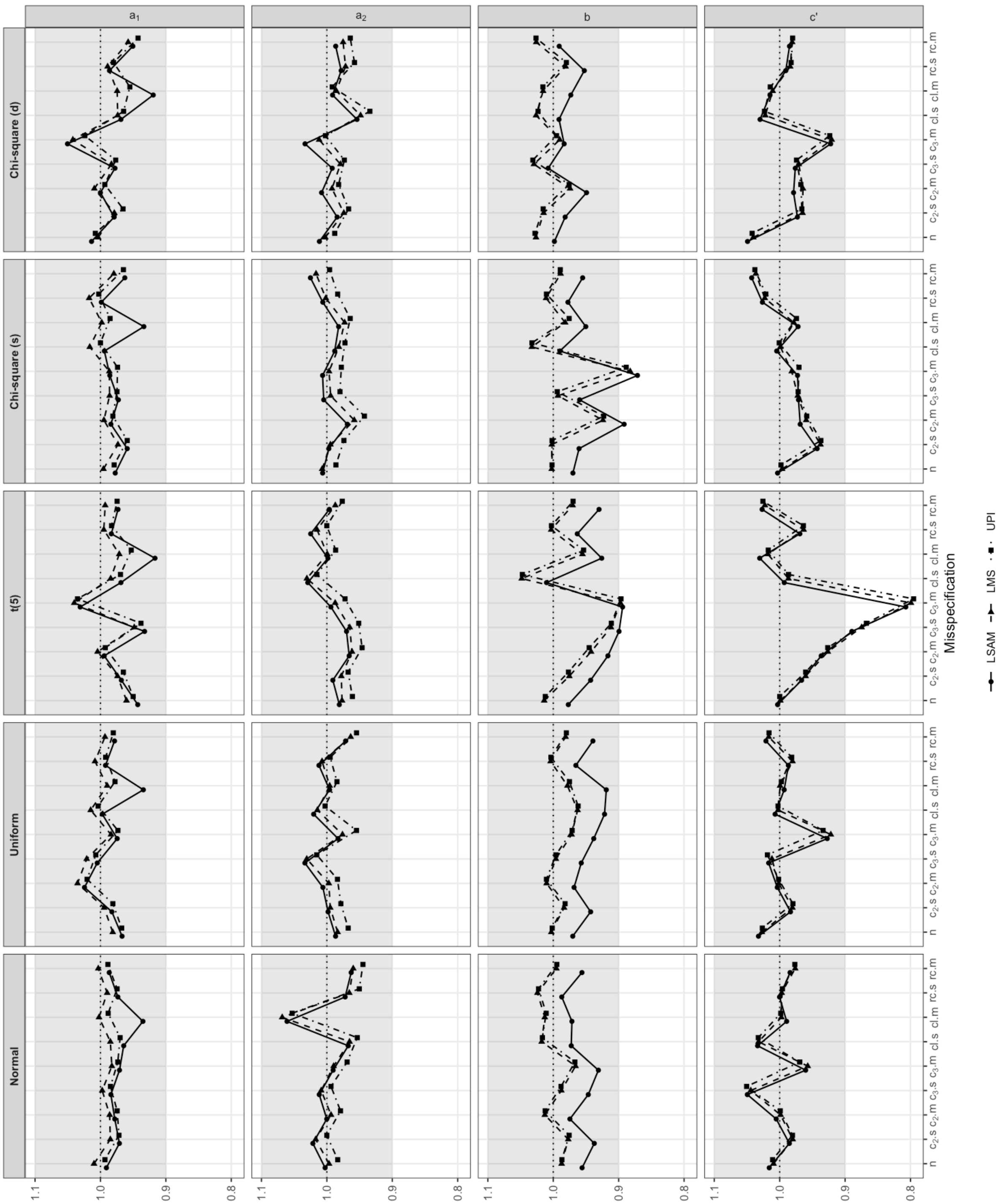
Coverage of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, N = 500, and a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 14

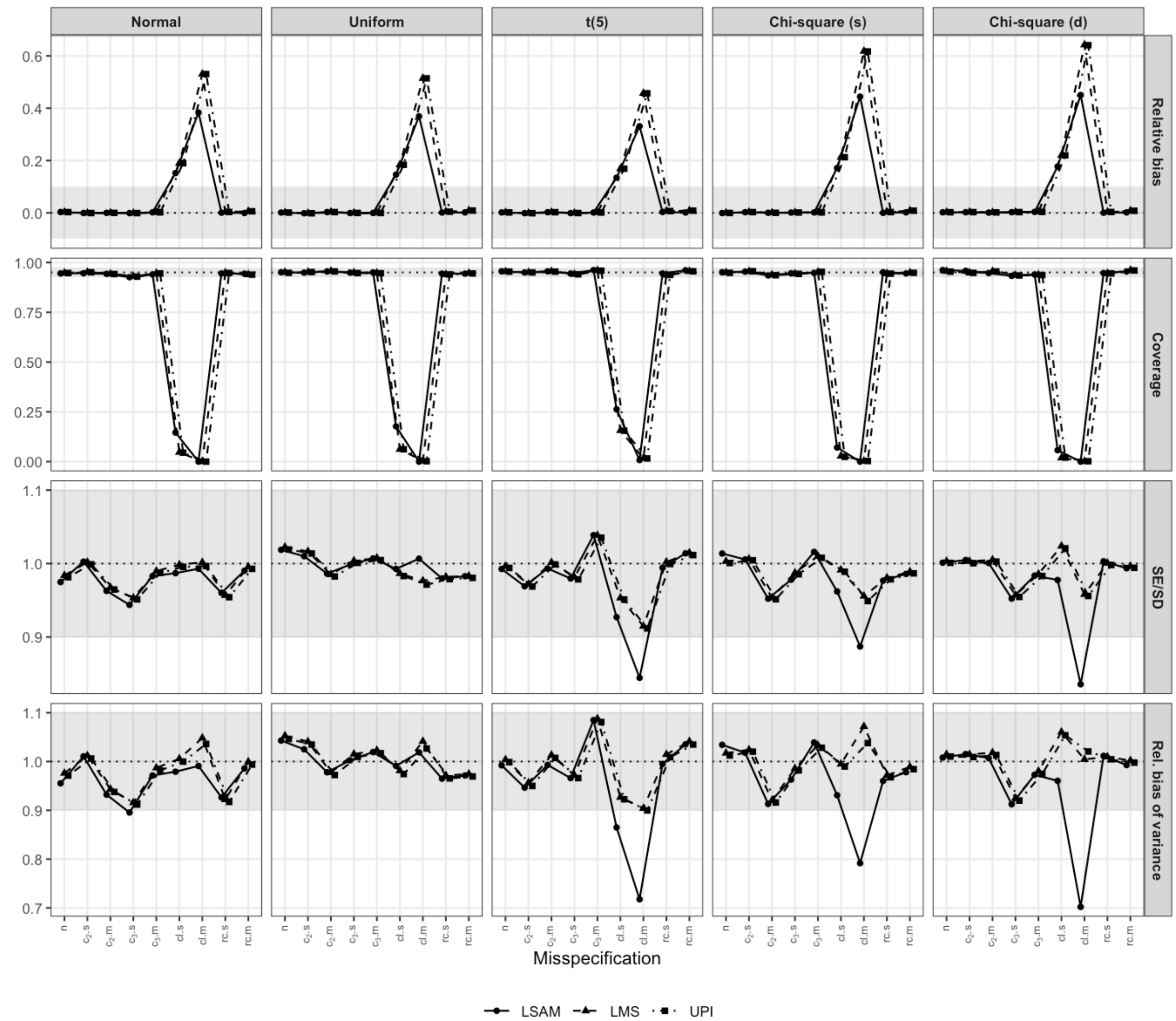
SE/SD of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, N = 500, and a_3 = 0.2



Note. The grey band marks the acceptable region for the measure. The MCSE is not computed for the SE/SD ratio.

Figure 15

Performance of the Cross-Loaded Mediator Indicator Loading (m_3) Across All Measures, Distributions, and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$



Note. The grey band in each panel marks the acceptable region for that measure.

Appendix

Calibration and Data Generation

This appendix gives the formulas behind the data-generating mechanism. The full derivations, with worked numerical examples and even more details, are in the project repository. The calibration that fixes the variance parameters is given first, followed by the three-stage data generation that uses them.

Calibration

The structural disturbance variances and indicator error variances were fixed by the following closed-form calibration, computed once under the normal baseline.

Structural residual variances

The disturbance variances ψ_M, ψ_Y are fixed so that each structural equation explains a fraction $R^2 = 0.30$ of its outcome's variance at the central interaction level $a_3 = 0.2$ under the normal baseline. Writing the systematic part of M as $S_M = a_1 X + a_2 W + a_3(XW)$, its variance needs only second moments,

$$\mathrm{Var}(S_M) = a_1^2 + a_2^2 + 2a_1 a_2 \rho + a_3^2(1 + \rho^2),$$

and the disturbance is the remainder that brings the signal share to R^2,

$$\psi_M = \mathrm{Var}(S_M)\,\frac{1 - R^2}{R^2}, \qquad \psi_Y = \mathrm{Var}(bM + c'X)\,\frac{1 - R^2}{R^2},$$

with $\mathrm{Var}(bM + c'X) = b^2\,\mathrm{Var}(M) + c'^2 + 2bc'\,(a_1 + a_2\rho)$. Fixed at these values, the population variances of M and Y are closed-form functions of the parameters,

$$\mathrm{Var}(M) = \mathrm{Var}(S_M) + \psi_M, \qquad \mathrm{Var}(Y) = b^2\,\mathrm{Var}(M) + c'^2 + 2bc'\,(a_1 + a_2\rho) + \psi_Y,$$

evaluated once under the normal baseline and held fixed across distributions, so that R^2 and reliability drift only through the design factors.

Indicator reliabilities

To hit a target reliability $\omega \in \{0.5, 0.7\}$, the error variance of indicator j is set from the latent variance $V = \mathrm{Var}(\eta)$,

$$\theta_{\eta j} = \lambda_j^2 V \frac{1-\omega}{\omega}.$$

For X, W the variance is exactly 1; for M, Y it is the closed form above, fixed at the normal baseline, so the realized reliability drifts mildly across distributions.

Calibrating the misspecification magnitudes

The *small* and *medium* labels are standardized coefficients, $t \in \{0.25, 0.50\}$. For an omitted predictor p added to a baseline outcome y_0, the magnitude is the fully standardized regression coefficient of p,

$$t = \frac{\beta \ \mathrm{sd}(p)}{\mathrm{sd}(y_0 + \beta p)}.$$

Squaring and substituting $\mathrm{Var}(y_0 + \beta p) = \mathrm{Var}(y_0) + \beta^2 \mathrm{Var}(p) + 2\beta \mathrm{Cov}(p, y_0)$ yields a quadratic $A\beta^2 + B\beta + C = 0$ with

$$A = \mathrm{Var}(p)\,(1-t^2), \quad B = -2t^2 \mathrm{Cov}(p, y_0), \quad C = -t^2 \mathrm{Var}(y_0).$$

Since $A > 0$ and $C < 0$ the roots have opposite sign; the positive omitted path is the positive root,

$$\beta = \frac{t^2 \mathrm{Cov}(p, y_0) + \sqrt{t^4 \mathrm{Cov}(p, y_0)^2 + t^2(1-t^2)\ \mathrm{Var}(p)\ \mathrm{Var}(y_0)}}{\mathrm{Var}(p)\,(1-t^2)}.$$

The three moments are estimated from one large ($n = 10^6$) sample per distribution and a_3 level. The predictor p and baseline y_0 differ by misspecification,

$$
\begin{aligned}
c_2: &\quad p = W, &\quad y_0 &= bM + c'X + \varepsilon_Y,\\
c_3: &\quad p = XW, &\quad y_0 &= bM + c'X + \varepsilon_Y,\\
\texttt{cl}: &\quad p = X, &\quad y_0 &= m_3 = \lambda_3 M + \delta_{m_3}.
\end{aligned}
$$

c_2 and c_3 are reliability independent; `cl` depends on ω because its baseline m_3 carries measurement noise. `rc` never enters the calibration: it is a residual correlation and equals its target in every cell.

Holding Y reliability fixed under c_2 and c_3

Adding an omitted structural path inflates $\mathrm{Var}(Y)$ by $\kappa = \mathrm{Var}(y_0 + \beta p)/\mathrm{Var}(y_0)$, which would raise Y's indicator reliability as a side effect. Scaling the Y-indicator noise by the same κ makes signal and noise move together, so κ cancels and the reliability returns to its no-misspecification value,

$$
\omega_j = \frac{\kappa\lambda_j^2 V}{\kappa\lambda_j^2 V + \kappa\theta_j} = \frac{\lambda_j^2 V}{\lambda_j^2 V + \theta_j}.
$$

The latent effect size is untouched; only the Y indicators are made correspondingly noisier. The measurement level misspecifications `cl` and `rc` act on the indicators directly and are not compensated, since that contamination is itself the misspecification.

Data generation

Using the calibrated variances above, data were generated in three stages: the correlated exogenous predictors, the dependent latent variables through the structural model, and the observed indicators through the measurement model.

Stage I: Exogenous predictors

The correlated predictors (X, W) were generated with the vine-to-anything (VITA) method (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022), as implemented in the *covsim* package, with sampling via *rvinecopulib* (Nagler & Vatter, 2025). They were specified to have mean 0, variance 1, and the covariance structure

$$\Phi = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.3 \\ 0.3 & 1 \end{bmatrix},$$

so that the conditions differ only in marginal shape. Under the normal condition, the predictors had Gaussian marginals with the specified covariance structure. Under the uniform condition, they had symmetric uniform marginals $\mathrm{U}(-\sqrt{3}, \sqrt{3})$, which have mean 0 and variance 1. Under the t_5 condition, they had Student t marginals with five degrees of freedom, rescaled by $\sqrt{5/3}$ to unit variance. The two χ^2 conditions used χ^2_2 marginals, centered by their mean (2) and scaled by their standard deviation (2) to mean 0 and variance 1. In the same-direction condition both predictors are right-skewed, whereas in the opposite-direction condition W is negated so that X and W are skewed in opposite directions, with the sampled correlation sign flipped to restore $\mathrm{Cor}(X, W) = \rho$. In every condition, VITA constructed a vine copula to impose Φ while preserving the marginals.

Stage II: Structural model

The dependent latent variables were generated from the structural equations

$$M = a_1 X + a_2 W + a_3 (X \times W) + \zeta_M, \qquad Y = bM + c'X + \zeta_Y,$$

with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$. The disturbance variances ψ_M, ψ_Y take the calibrated values so that each equation explains $R^2 \approx 0.30$ of its outcome's variance at the central interaction level under the normal baseline.

Stage III: Measurement model

Each latent $\eta \in \{X, W, M, Y\}$ was measured by four indicators,

$$z_{\eta j} = \lambda_j \, \eta + \delta_{\eta j}, \qquad \lambda = (1, 0.8, 0.7, 0.6),$$

with independent Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$. The error variances $\theta_{\eta j}$ take the calibrated values that achieve the target reliability $\omega \in \{0.5, 0.7\}$. Only the exogenous predictors varied in distribution. The structural disturbances and measurement errors were Gaussian in every condition.